9

Estimating Equations and Maximum Likelihood

Many estimators in statistics are specified implicitly as solutions to equations or as values maximizing some function. In this chapter we study why these methods work and learn ways to approximate distributions. Although we focus on methods for i.i.d. observations, many of the ideas can be extended. Results for stationary time series are sketched in Section 9.9.

A first example, introduced in Section 8.3, concerns maximum likelihood estimation. The maximum likelihood estimator $\hat{\theta}$ maximizes the likelihood function $L(\cdot)$ or log-likelihood $l(\cdot) = \log L(\cdot)$. And if l is differentiable and the maximum occurs in the interior of the parameter space, then $\hat{\theta}$ solves $\nabla l(\theta) = 0$. Method of moments estimators, considered in Problem 9.2, provide a second example. If $X_1, \dots, X_n$ are i.i.d. observations with average $\overline{X}$, and if $\mu(\theta) = E_\theta X_i$, then the method of moments estimator of θ solves $\mu(\theta) = \overline{X}$. A final example would be M-estimators, considered in Section 9.8.

9.1 Weak Law for Random Functions[1]

In this section we develop a weak law of large numbers for averages of random functions. This is used in the rest of the chapter to establish consistency and asymptotic normality of maximum likelihood and other estimators.

Let $X_1, X_2, \dots$ be i.i.d., let K be a compact set in $\mathbb{R}^p$, and define

$$W_i(t) = h(t, X_i), \qquad t \in K,$$

where $h(t, x)$ is a continuous function of t for all x. Then $W_1, W_2, \dots$ are i.i.d. random functions taking values in $C(K)$, the space of continuous functions on K.

Functions in $C(K)$ behave in many ways like vectors. They can be added, subtracted, and multiplied by constants, with these operations satisfying the

[1] The theory developed in this section is fairly technical, but uniform convergence is important for applications developed in later sections.

R.W. Keener, *Theoretical Statistics: Topics for a Core Course*, Springer Texts in Statistics, DOI 10.1007/978-0-387-93839-4_9,

usual properties. Sets with these properties are called *linear spaces*. In addition, notions of convergence can be introduced for functions in $C(K)$. There are various possibilities. The one we use in this section is based on a notion of length. For $w \in C(K)$ define

$$\|w\|_\infty = \sup_{t \in K} |w(t)|,$$

called the *supremum norm* of w. Functions w_n converge to w in this norm if $\|w_n - w\|_\infty \to 0$. With this norm, $C(K)$ is complete (all Cauchy sequences converge), and a complete linear space with a norm is called a *Banach space*.

A final nice property of $C(K)$ is separability. A subset of some set is called *dense* if every element in the set is arbitrarily close to some point in the subset. For instance, the rational numbers are a dense subset of $\mathbb{R}$ because there are rational numbers arbitrarily close to any real number $x \in \mathbb{R}$. A space is *separable* if it has a countable dense subset. We state the law of large numbers in this section for i.i.d. random functions in $C(K)$, but the result also holds for i.i.d. random elements in an arbitrary separable Banach space.[2]

Lemma 9.1. *Let W be a random function in $C(K)$ and define*

$$\mu(t) = EW(t), \qquad t \in K.$$

(This function μ is called the mean *of W.) If $E\|W\|_\infty < \infty$, then μ is continuous. Also,*

$$\sup_{t \in K} E \sup_{s:\|s-t\|<\epsilon} |W(s) - W(t)| \to 0$$

as $\epsilon \downarrow 0$.

Proof. Let t_n, $n \geq 1$, be a sequence of constants in K converging to t. Because W is continuous, the random variables $W(t_n)$ converge to $W(t)$ as $n \to \infty$. They are also dominated by $\|W\|_\infty$, which has a finite expectation. Thus

$$\mu(t_n) = EW(t_n) \to EW(t) = \mu(t)$$

as $n \to \infty$ by dominated convergence, and μ is continuous.

For the second part, define

$$M_\epsilon(t) = \sup_{s:\|s-t\|<\epsilon} |W(s) - W(t)|,$$

[2] As usual, we are not giving much attention to issues of measurability, and the notion of what we mean by a "random function" is a bit vague. To be more specific, define $B_a(w) = \{f \in C(K) : \|f - w\|_\infty < a\}$, called the open ball with radius a centered at w. The Borel σ-field $\mathcal{B}$ can then be defined as the smallest σ-field that contains all open balls. If probability is defined on a measurable space $(\mathcal{X}, \mathcal{A})$, then $W : \mathcal{X} \to C(K)$ is *measurable* and would be called a random function if $W^{-1}(B) \in \mathcal{A}$ for any Borel set $B \in \mathcal{B}$. Aside from defining Borel sets using open balls instead of intervals, this definition is essentially the same as the definition of measurability for random variables given in Definition 1.7.

and let λ_ϵ be the mean of M_ϵ,

$$\lambda_\epsilon(t) = EM_\epsilon(t).$$

Because W is continuous, M_ϵ is continuous. Also, since $|M_\epsilon(t)| \le 2\|W\|_\infty$, $E\|M_\epsilon\|_\infty < \infty$, and by the first part of this theorem, λ_ϵ is continuous. By continuity, $M_\epsilon(t) \to 0$ as $\epsilon \to 0$, and by dominated convergence, $\lambda_\epsilon(t) \to 0$ as $\epsilon \to 0$. Since the functions λ_ϵ are decreasing as $\epsilon \downarrow 0$, by Dini's theorem (Theorem A.5) the convergence is uniform, that is, $\sup_{t\in K} \lambda_\epsilon(t) \to 0$ as $\epsilon \downarrow 0$. $\square$

Theorem 9.2. *Let $W, W_1, W_2, \ldots$ be i.i.d. random functions in $C(K)$, K compact, with mean μ and $E\|W\|_\infty < \infty$, and let $\overline{W}_n = (W_1 + \cdots + W_n)/n$. Then*

$$\|\overline{W}_n - \mu\|_\infty \xrightarrow{p} 0$$

as $n \to \infty$.

By the weak law of large numbers, for any $t \in K$, $\overline{W}_n(t) \xrightarrow{p} \mu(t)$. But the theorem is stronger, asserting that this convergence holds with uniformity in t.

Proof. Fix $\epsilon > 0$. For notation, let

$$M_{\delta,j}(t) = \sup_{s:\|s-t\|<\delta} \bigl|W_j(s) - W_j(t)\bigr|$$

with mean $\lambda_\delta(t)$. Choose δ using the second assertion of Lemma 9.1 so that

$$\lambda_\delta(t) = E \sup_{s:\|s-t\|<\delta} |W(s) - W(t)| < \epsilon, \qquad \forall t \in K,$$

and note that with this choice of δ, if $\|t - s\| < \delta$, then

$$|\mu(t) - \mu(s)| = \bigl|E\bigl[W(t) - W(s)\bigr]\bigr| \le E\bigl|W(t) - W(s)\bigr| \le \epsilon.$$

Let $B_\delta(t) = \bigl\{s : \|s - t\| < \delta\bigr\}$, the open ball with radius δ about t. Since K is compact, the open sets $B_\delta(t)$, $t \in K$, covering K have a finite subcover $O_i = B_\delta(t_i)$, $i = 1, \ldots, m$. Then

$$\begin{aligned}
&\|\overline{W}_n - \mu\|_\infty \\
&\qquad = \max_{i=1,\ldots,m} \sup_{t\in O_i} \bigl|\overline{W}_n(t) - \mu(t)\bigr| \\
&\qquad \le \max_i \sup_{t\in O_i} \Bigl[\bigl|\overline{W}_n(t) - \overline{W}_n(t_i)\bigr| + \bigl|\overline{W}_n(t_i) - \mu(t_i)\bigr| + \bigl|\mu(t_i) - \mu(t)\bigr|\Bigr] \\
&\qquad \le \max_i \sup_{t\in O_i} \bigl|\overline{W}_n(t) - \overline{W}_n(t_i)\bigr| + \max_i \bigl|\overline{W}_n(t_i) - \mu(t_i)\bigr| + \epsilon.
\end{aligned}$$

Now

$$
\begin{aligned}
\sup_{t\in O_i}\left|\overline{W}_n(t)-\overline{W}_n(t_i)\right| &= \frac{1}{n}\sup_{t\in O_i}\left|\sum_{j=1}^{n}\left[W_j(t)-W_j(t_i)\right]\right| \\
&\le \frac{1}{n}\sum_{j=1}^{n}\sup_{t\in O_i}\left|W_j(t)-W_j(t_i)\right| \\
&= \frac{1}{n}\sum_{j=1}^{n} M_{\delta,j}(t_i) \overset{\text{def}}{=} \overline{M}_{\delta,n}(t_i).
\end{aligned}
$$

By the law of large numbers,

$$
\overline{M}_{\delta,n}(t_i) \xrightarrow{p} \lambda_\delta(t_i) < \epsilon.
$$

Using these bounds,

$$
\|\overline{W}_n-\mu\|_\infty < 2\epsilon + \max_i\bigl(\overline{M}_{\delta,n}(t_i)-\lambda_\delta(t_i)\bigr) + \max_i\bigl|\overline{W}_n(t_i)-\mu(t_i)\bigr|.
$$

The two maximums in this equation both converge to zero in probability and using this it is easy to argue that $P\bigl(\|\overline{W}_n-\mu\|_\infty > 3\epsilon\bigr) \to 0$ as $n\to\infty$. ☐

Remark 9.3. The same proof coupled with the strong law of large numbers, stated in Section 8.7, shows that $\|\overline{W}_n-\mu\|_\infty \to 0$ almost surely.

The following result shows the usefulness of uniform convergence. None of the conclusions follow from pointwise convergence in probability.

Theorem 9.4. *Let G_n, $n\ge 1$, be random functions in $C(K)$, K compact, and suppose $\|G_n-g\|_\infty \xrightarrow{p} 0$ with g a nonrandom function in $C(K)$.*

1. *If t_n, $n\ge 1$, are random variables converging in probability to a constant $t^*\in K$, $t_n \xrightarrow{p} t^*$, then $G_n(t_n)\xrightarrow{p} g(t^*)$.*
2. *If g achieves its maximum at a unique value t^*, and if t_n are random variables maximizing G_n, so that*

$$
G_n(t_n) = \sup_{t\in K} G_n(t),
$$

then $t_n \xrightarrow{p} t^*$.

3. *If $K\subset\mathbb{R}$ and $g(t)=0$ has a unique solution t^*, and if t_n are random variables solving $G_n(t_n)=0$, then $t_n\xrightarrow{p} t^*$.*

Proof. For the first assertion, since

$$
\begin{aligned}
\bigl|G_n(t_n)-g(t^*)\bigr| &\le \bigl|G_n(t_n)-g(t_n)\bigr| + \bigl|g(t_n)-g(t^*)\bigr| \\
&\le \|G_n-g\|_\infty + \bigl|g(t_n)-g(t^*)\bigr|,
\end{aligned}
$$

and since $g(t_n)\xrightarrow{p} g(t^*)$,

$$
\begin{aligned}
P\big(\big|G_n(t_n) - g(t^*)\big| > \epsilon\big) \\
&\le P\left(\|G_n - g\|_\infty + \big|g(t_n) - g(t^*)\big| > \epsilon\right) \\
&\le P\left(\|G_n - g\|_\infty > \epsilon/2\right) + P\left(\big|g(t_n) - g(t^*)\big| > \epsilon/2\right) \\
&\to 0.
\end{aligned}
$$

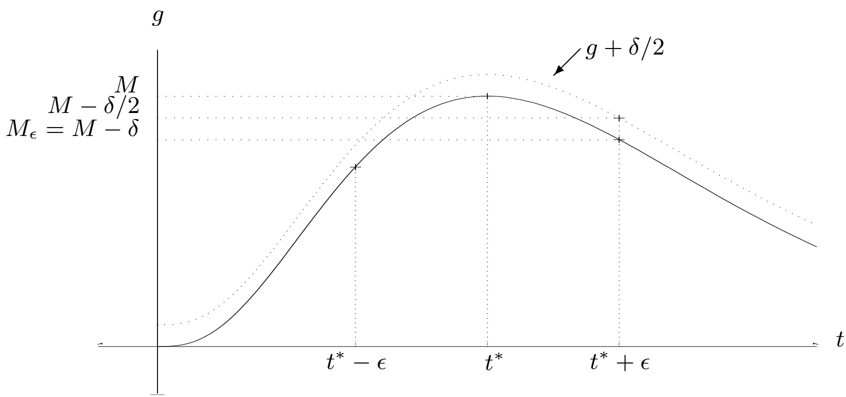

Fig. 9.1. g and $g + \delta/2$.

For the second assertion, fix ϵ and let $K_\epsilon = K - B_\epsilon(t^*)$. This set is compact; it is bounded because K is bounded, and it is closed because it is the intersection of two closed sets, K and the complement of $B_\epsilon(t^*)$. Let $M = g(t^*) = \sup_K g$ and let $M_\epsilon = \sup_{K_\epsilon} g$. Since K_ϵ is compact, $M_\epsilon = g(t_\epsilon^*)$ for some $t_\epsilon^* \in K_\epsilon$, and since g has a unique maximum over K, $M_\epsilon < M$. Define $\delta = M - M_\epsilon > 0$. See Figure 9.1. Suppose $\|G_n - g\|_\infty < \delta/2$. Then

$$\sup_{K_\epsilon} G_n < \sup_{K_\epsilon} g + \frac{\delta}{2} = M - \frac{\delta}{2}$$

and

$$\sup_K G_n \ge G_n(t^*) > g(t^*) - \frac{\delta}{2} = M - \frac{\delta}{2},$$

and t_n must lie in $B_\epsilon(t^*)$. Thus

$$P\big(\|G_n - g\|_\infty < \delta/2\big) \le P\big(\|t_n - t^*\| < \epsilon\big).$$

Taking complements,

$$P\big(\|t_n - t^*\| \ge \epsilon\big) \le P\big(\|G_n - g\|_\infty \ge \delta/2\big) \to 0,$$

and so $t_n \xrightarrow{p} t^*$. The third assertion in the theorem can be established in a similar fashion. $\square$

Remark 9.5. The law of large numbers and the first and third assertions in Theorem 9.4 can be easily extended to multivariate situations where the random functions are vector-valued, mapping a compact set K into $\mathbb{R}^p$.

Remark 9.6. In the approach to the weak law here, continuity plays a key role in proving uniform convergence. Uniform convergence without continuity is also possible. One important result concerns empirical distribution functions. If $X_1, \dots, X_n$ are i.i.d., then a natural estimator for the common cumulative distribution function F would be the *empirical cumulative distribution function* $\hat{F}_n$, defined as

$$\hat{F}_n(x) = \frac{1}{n}\#\{i \le n : X_i \le x\}, \qquad x \in \mathbb{R}.$$

The Glivenko–Cantelli theorem asserts that $\|\hat{F}_n - F\|_\infty \xrightarrow{p} 0$ as $n \to \infty$. In the proof of this result, monotonicity replaces continuity as the key regularity used to establish uniform convergence.

9.2 Consistency of the Maximum Likelihood Estimator

For this section let $X, X_1, X_2, \dots$ be i.i.d. with common density f_θ, $\theta \in \Omega$, and let l_n be the log-likelihood function for the first n observations:

$$l_n(\omega) = \log \prod_{i=1}^n f_\omega(X_i) = \sum_{i=1}^n \log f_\omega(X_i).$$

(We use ω as the dummy argument here, reserving θ to represent the true value of the unknown parameter in the sequel.) Then the maximum likelihood estimator $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$ from the first n observations will maximize l_n. For regularity, assume $f_\theta(x)$ is continuous in θ.

Definition 9.7. *The* Kullback–Leibler information *is defined as*

$$I(\theta, \omega) = E_\theta \log\bigl[f_\theta(X)/f_\omega(X)\bigr].$$

It can be viewed as a measure of the information discriminating between θ and ω when θ is the true value of the unknown parameter.

Lemma 9.8. *If $P_\theta \ne P_\omega$, then $I(\theta, \omega) > 0$.*

Proof. By Jensen's inequality,

$$\begin{aligned}
-I(\theta, \omega) &= E_\theta \log\bigl[f_\omega(X)/f_\theta(X)\bigr] \\
&\le \log E_\theta \bigl[f_\omega(X)/f_\theta(X)\bigr] \\
&= \log \int_{f_\theta > 0} \frac{f_\omega(x)}{f_\theta(x)} f_\theta(x)\, d\mu(x) \\
&\le \log 1 \\
&= 0.
\end{aligned}$$

Strict equality will occur only if $f_\omega(X)/f_\theta(X)$ is constant a.e. But then the densities will be proportional and hence equal a.e., and P_θ and P_ω will be the same. $\square$

The next result gives consistency for the maximum likelihood estimator when Ω is compact. The result following is an extension when $\Omega = \mathbb{R}^p$. Define

$$W(\omega) = \log\left[\frac{f_\omega(X)}{f_\theta(X)}\right].$$

Theorem 9.9. *If Ω is compact, $E_\theta\|W\|_\infty < \infty$, $f_\omega(x)$ is a continuous function of ω for a.e. x, and $P_\omega \neq P_\theta$ for all $\omega \neq \theta$, then under P_θ, $\hat{\theta}_n \xrightarrow{p} \theta$.*

Proof. If $W_i(\omega) = \log\bigl(f_\omega(X_i)/f_\theta(X_i)\bigr)$, then under P_θ, $W_1, W_2, \dots$ are i.i.d. random functions in $C(\Omega)$ with mean $\mu(\omega) = -I(\theta, \omega)$. Note that $\mu(\theta) = 0$ and $\mu(\omega) < 0$ for $\omega \neq \theta$ by Lemma 9.8, and so μ has a unique maximum at θ. Since

$$\overline{W}_n(\omega) = \frac{1}{n}\sum_{j=1}^n W_i(\omega) = \frac{l_n(\omega) - l_n(\theta)}{n},$$

$\hat{\theta}_n$ maximizes $\overline{W}_n$. By Theorem 9.2, $\|\overline{W}_n - \mu\|_\infty \to 0$, and the result follows from the second assertion of Theorem 9.4. $\square$

Remark 9.10. The argument used to prove consistency here is based on the proof in Wald (1949). In this paper, the one-sided condition that $E_\theta \sup_\Omega W < \infty$ replaces $E_\theta\|W\|_\infty < \infty$. Inspecting the proof here, it is not hard to see that Wald's weaker condition is sufficient.

Theorem 9.11. *Suppose $\Omega = \mathbb{R}^p$, $f_\omega(x)$ is a continuous function of ω for a.e. x, $P_\omega \neq P_\theta$ for all $\omega \neq \theta$, and $f_\omega(x) \to 0$ as $\omega \to \infty$. If $E_\theta\|1_K W\|_\infty < \infty$ for any compact set $K \subset \mathbb{R}^p$, and if $E_\theta \sup_{\|\omega\|>a} W(\omega) < \infty$ for some $a > 0$, then under P_θ, $\hat{\theta}_n \xrightarrow{p} \theta$.*

Proof. Since $f_\omega(x) \to 0$ as $\omega \to \infty$, if $f_\theta(X) > 0$,

$$\sup_{\|\omega\|>b} W(\omega) \to -\infty$$

as $b \to \infty$. By a dominated convergence argument the expectation of this variable will tend to $-\infty$ as $b \to \infty$, and we can choose b so that

$$E_\theta \sup_{\|\omega\|>b} W(\omega) < 0.$$

Note that b must exceed $\|\theta\|$, because $W(\theta) = 0$. Since

$$\sup_{\|\omega\|>b} \overline{W}_n(\omega) \le \frac{1}{n}\sum_{j=1}^n \sup_{\|\omega\|>b} W_i(\omega) \xrightarrow{p} E_\theta \sup_{\|\omega\|>b} W(\omega),$$

$$P_\theta\Big(\sup_{\|\omega\|>b} \overline{W}_n(\omega) \ge 0\Big) \to 0,$$

as $n \to \infty$. Let K be the closed ball of radius b, and let $\tilde\theta_n$ be variables maximizing $\overline{W}_n$ over K.[3] By Theorem 9.9, $\tilde\theta_n \xrightarrow{p} \theta$. Since $\hat\theta_n$ must lie in K whenever

$$\sup_{\|\omega\|>b} \overline{W}_n(\omega) < \overline{W}_n(\theta) = 0,$$

$P_\theta(\hat\theta_n = \tilde\theta_n) \to 1$. It then follows that $\hat\theta_n \xrightarrow{p} \theta$. ☐

Remark 9.12. A similar result can be obtained when Ω is an arbitrary open set. The corresponding conditions would be that $f_\omega(x) \to 0$ as ω approaches the boundary of Ω, and that $E_\theta \sup_{\omega\in K^c} W(\omega) < \infty$ for some compact set K. Although conditions for consistency are fairly mild, counterexamples are possible when they fail. Problem 9.4 provides one example.

Example 9.13. Suppose we have a location family with densities $f_\theta(x) = g(x-\theta)$, $\theta \in \mathbb{R}$, and that

1. g is continuous and bounded, so $\sup_{x\in\mathbb{R}} g(x) = K < \infty$,
2. $g(x) \to 0$ as $x \to \pm\infty$, and
3. $\int |\log g(x)| g(x)\,dx < \infty$.

Then

$$\begin{aligned}
E_\theta \sup_{\omega\in\mathbb{R}} W(\omega) &= E_\theta \sup_{\omega\in\mathbb{R}} \log \frac{g(X-\omega)}{g(X-\theta)} \\
&= \log K - E_\theta \log g(X-\theta) \\
&= \log K - \int [\log g(x)] g(x)\,dx \\
&< \infty.
\end{aligned}$$

Hence $\hat\theta_n$ is consistent by the one-sided adaptation of our consistency theorems mentioned in Remark 9.10. The third condition here is not very stringent; it holds for most densities, including the Cauchy and other t-densities, that decay algebraically near infinity.

9.3 Limiting Distribution for the MLE

Theorem 9.14. *Assume:*

1. *Variables $X, X_1, X_2, \ldots$ are i.i.d. with common density f_θ, $\theta \in \Omega \subset \mathbb{R}$.*

[3] To be careful, as we define $\tilde\theta_n$, we should also insist that $\tilde\theta_n = \hat\theta_n$ whenever $\hat\theta_n \in K$, to cover cases with multiple maxima.

2. *The set* $A = \{x : f_\theta(x) > 0\}$ *is independent of* θ.
3. *For every* $x \in A$, $\partial^2 f_\theta(x)/\partial\theta^2$ *exists and is continuous in* θ.
4. *Let* $W(\theta) = \log f_\theta(X)$. *The Fisher information* $I(\theta)$ *from a single observation exists, is finite, and can be found using either*

$$I(\theta) = E_\theta W'(\theta)^2 \ \text{ or } \ I(\theta) = -E_\theta W''(\theta).$$

   Also,

$$E_\theta W'(\theta) = 0.$$

5. *For every* θ *in the interior of* Ω *there exists* $\epsilon > 0$ *such that*

$$E_\theta \|1_{[\theta-\epsilon,\theta+\epsilon]} W''\|_\infty < \infty.$$

6. *The maximum likelihood estimator* $\hat\theta_n$ *is consistent.*

Then for any θ *in the interior of* Ω,

$$\sqrt{n}(\hat\theta_n - \theta) \Rightarrow N\bigl(0, 1/I(\theta)\bigr)$$

under P_θ *as* $n \to \infty$.

The assumptions in this theorem are fairly mild, although similar results, such as those in Chapter 16, are possible under weaker conditions. Assumption 2 usually precludes families of uniform distributions or truncated families. Assumptions 3 and 4 are the same as assumptions discussed for the Cramér–Rao bound, and Assumption 5 strengthens 4. Concerning the final assumption, for the proof $\hat\theta_n$ needs to be consistent, but it is not essential that it maximizes the likelihood. What matters is that $\sqrt{n}\overline{W}'_n(\hat\theta_n) \xrightarrow{P_\theta} 0$. In regular cases this will hold for Bayes estimators. There may also be models satisfying the other assumptions for this theorem in which the maximum likelihood estimator does not exist or is not consistent. In these examples there is often a consistent $\hat\theta_n$ solving $\overline{W}'_n(\hat\theta_n) = 0$, with this consistent root of the likelihood equation asymptotically normal.

The following technical lemma shows that, when proving convergence in distribution, we only need consider what happens on a sequence of events with probabilities converging to one.

Lemma 9.15. *Suppose* $Y_n \Rightarrow Y$, *and* $P(B_n) \to 1$ *as* $n \to \infty$. *Then for arbitrary random variables* Z_n, $n \ge 1$,

$$Y_n 1_{B_n} + Z_n 1_{B_n^c} \Rightarrow Y$$

as $n \to \infty$.

Proof. For any $\epsilon > 0$,

$$P\bigl(|Z_n 1_{B_n^c}| > \epsilon\bigr) \le P(B_n^c) = 1 - P(B_n) \to 0$$

as $n \to \infty$. So $Z_n 1_{B_n^c} \xrightarrow{p} 0$ as $n \to \infty$. Also,

$$P\bigl(|1_{B_n} - 1| > \epsilon\bigr) \le P(B_n^c) = 1 - P(B_n) \to 0$$

as $n \to \infty$, and so $1_{B_n} \xrightarrow{p} 1$ as $n \to \infty$. With these observations, the lemma now follows from Theorem 8.13. ☐

Proof of Theorem 9.14. Choose $\epsilon > 0$ using Assumption 5 small enough that $[\theta - \epsilon, \theta + \epsilon] \subset \Omega^0$ and $E_\theta \| 1_{[\theta-\epsilon,\theta+\epsilon]} W'' \|_\infty < \infty$, and let B_n be the event $\hat{\theta}_n \in [\theta - \epsilon, \theta + \epsilon]$. Because $\hat{\theta}_n$ is consistent, $P_\theta(B_n) \to 1$, and since $\hat{\theta}_n$ maximizes $n\overline{W}_n(\cdot) = l_n(\cdot)$, on B_n we have $\overline{W}'_n(\hat{\theta}_n) = 0$. Taylor expansion of $\overline{W}'_n$ about θ gives

$$\overline{W}'_n(\hat{\theta}_n) = \overline{W}'_n(\theta) + \overline{W}''_n(\tilde{\theta}_n)(\hat{\theta}_n - \theta),$$

where $\tilde{\theta}_n$ is an intermediate value between $\hat{\theta}_n$ and θ. Setting the left-hand side of this equation to zero and solving, on B_n,

$$\sqrt{n}(\hat{\theta}_n - \theta) = \frac{\sqrt{n}\overline{W}'_n(\theta)}{-\overline{W}''_n(\tilde{\theta}_n)}. \tag{9.1}$$

By Assumption 4, the variables averaged in $\overline{W}'_n(\theta)$ are i.i.d., mean zero, with variance $I(\theta)$. By the central limit theorem,

$$\sqrt{n}\,\overline{W}'_n(\theta) \Rightarrow Z \sim N\bigl(0, I(\theta)\bigr).$$

Turning to the denominator, since $|\tilde{\theta}_n - \theta| \le |\hat{\theta}_n - \theta|$, at least on B_n, and $\hat{\theta}_n$ is consistent, $\tilde{\theta}_n \xrightarrow{p} \theta$. By Theorem 9.2,

$$\| 1_{[\theta-\epsilon,\theta+\epsilon]}(\overline{W}''_n - \mu) \|_\infty \xrightarrow{p} 0,$$

where $\mu(\omega) = E_\theta W''(\omega)$, and so, by second assertion of Theorem 9.4,

$$\overline{W}''_n(\tilde{\theta}_n) \xrightarrow{p} \mu(\theta) = -I(\theta).$$

Since the behavior of $\hat{\theta}_n$ on B_n^c cannot affect convergence in distribution (by Lemma 9.15),

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow \frac{Z}{I(\theta)} \sim N\bigl(0, 1/I(\theta)\bigr),$$

as $n \to \infty$ by Theorem 8.13. ☐

Remark 9.16. The argument that

$$\overline{W}''_n(\tilde{\theta}_n) = \frac{1}{n} l''_n(\tilde{\theta}_n) \xrightarrow{p} -I(\theta)$$

holds for any variables $\tilde{\theta}_n$ converging to θ in probability. This is exploited later as we study asymptotic confidence intervals.

9.4 Confidence Intervals

A point estimator δ for an unknown parameter $g(\theta)$ provides no information about accuracy. Confidence intervals address this deficiency by seeking two statistics, δ_0 and δ_1, that bracket $g(\theta)$ with high probability.

Definition 9.17. *If δ_0 and δ_1 are statistics, then the random interval (δ_0, δ_1) is called a* $1-\alpha$ confidence interval *for $g(\theta)$ if*

$$P_\theta\big(g(\theta) \in (\delta_0, \delta_1)\big) \geq 1 - \alpha,$$

for all $\theta \in \Omega$. Also, a random set $S = S(X)$ constructed from data X is called a $1-\alpha$ confidence region *for $g(\theta)$ if*

$$P_\theta\big(g(\theta) \in S\big) \geq 1 - \alpha,$$

for all $\theta \in \Omega$.

Remark 9.18. In many examples, coverage probabilities equal $1-\alpha$ for all $\theta \in \Omega$, in which case the interval or region might be called an *exact* confidence interval or an exact confidence region.

Example 9.19. Let $X_1, \dots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$. Then from the results in Section 4.3, $\overline{X} = (X_1 + \cdots + X_n)/n$ and $S^2 = \sum_{i=1}^n (X_i - \overline{X})^2/(n-1)$ are independent, with $\overline{X} \sim N(\mu, \sigma^2/n)$ and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. Define

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

and

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

These variables Z and V are called *pivots*, since their distribution does not depend on the unknown parameters μ and σ^2. This idea is similar to ancillarity, but Z and V are not statistics since both variables depend explicitly on unknown parameters. Since Z and V are independent, the variable

$$T = \frac{Z}{\sqrt{V/(n-1)}} \tag{9.2}$$

is also a pivot. Its distribution is called the t-distribution on $n-1$ degrees of freedom, denoted $T \sim t_{n-1}$. The density for T is

$$f_T(x) = \frac{\Gamma\big((\nu+1)/2\big)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)(1 + x^2/\nu)^{(\nu+1)/2}}, \qquad x \in \mathbb{R},$$

where $\nu = n-1$, the number of degrees of freedom.

Pivots can be used to set confidence intervals. For $p \in (0,1)$, let $t_{p,\nu}$ denote the upper pth quantile for the t-distribution on ν degrees of freedom, so that

$$P\bigl(T \ge t_{p,\nu}\bigr) = \int_{t_{p,\nu}}^{\infty} f_T(x)\,dx = p.$$

By symmetry,

$$P\bigl(T \ge t_{\alpha/2,n-1}\bigr) = P\bigl(T \le -t_{\alpha/2,n-1}\bigr) = \alpha/2,$$

and so

$$P\bigl(-t_{\alpha/2,n-1} < T < t_{\alpha/2,n-1}\bigr) = 1-\alpha.$$

Now

$$T = \frac{Z}{\sqrt{V/(n-1)}} = \frac{\overline{X}-\mu}{S/\sqrt{n}},$$

and so

$$-t_{\alpha/2,n-1} < T < t_{\alpha/2,n-1}$$

if and only if

$$\frac{|\overline{X}-\mu|}{S/\sqrt{n}} < t_{\alpha/2,n-1}$$

if and only if

$$|\overline{X}-\mu| < t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$$

if and only if

$$\mu \in \left(\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right) \stackrel{\text{def}}{=} (\delta_0, \delta_1).$$

Thus for any $\theta = (\mu, \sigma^2)$,

$$P_\theta\bigl(\mu \in (\delta_0,\delta_1)\bigr) = 1-\alpha$$

and (δ_0, δ_1) is a $1-\alpha$ confidence interval for μ.

The pivot V can be used in a similar fashion to set confidence intervals for σ^2. Let $\chi^2_{p,\nu}$ denote the upper pth quantile for the chi-square distribution on ν degrees of freedom. Then

$$P\bigl(V \ge \chi^2_{\alpha/2,n-1}\bigr) = P\bigl(V \le \chi^2_{1-\alpha/2,n-1}\bigr) = \alpha/2,$$

and

$$\begin{aligned} 1-\alpha &= P_\theta\left(\chi^2_{1-\alpha/2,n-1} < V = \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2,n-1}\right) \\ &= P_\theta\left[\sigma^2 \in \left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right)\right]. \end{aligned}$$

Thus

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right)$$

is a $1-\alpha$ confidence interval for σ^2.

9.5 Asymptotic Confidence Intervals

Suppose the conditions of Theorem 9.14 hold, so that under P_θ,

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N\big(0, 1/I(\theta)\big)$$

as $n \to \infty$, where $\hat{\theta}_n$ is the maximum likelihood estimator of θ based on n observations. Multiplying by $\sqrt{I(\theta)}$, this implies

$$\sqrt{nI(\theta)}(\hat{\theta}_n - \theta) \Rightarrow N(0,1). \tag{9.3}$$

Since the limiting distribution here is independent of θ, $\sqrt{nI(\theta)}(\hat{\theta}_n - \theta)$ is called an *approximate pivot*. If we define $z_p = \Phi^{\leftarrow}(1-p)$, the upper pth quantile of $N(0,1)$, then

$$P_\theta\big(\sqrt{nI(\theta)}|\hat{\theta}_n - \theta| < z_{\alpha/2}\big) \to 1-\alpha$$

as $n \to \infty$. If we define the (random) set

$$S = \big\{\theta \in \Omega : \sqrt{nI(\theta)}|\hat{\theta}_n - \theta| < z_{\alpha/2}\big\}, \tag{9.4}$$

then $\theta \in S$ if and only if $\sqrt{nI(\theta)}|\hat{\theta}_n - \theta| < z_{\alpha/2}$, and so

$$P_\theta(\theta \in S) \to 1-\alpha$$

as $n \to \infty$. This set S is called a $1-\alpha$ *asymptotic confidence region* for θ.

Practical considerations may make the confidence region S in (9.4) undesirable. It need not be an interval, which may make the region hard to describe and difficult to interpret. Also, if the Fisher information $I(\cdot)$ is a complicated function, the inequalities defining the region may be difficult to solve. To avoid these troubles, note that if $I(\cdot)$ is continuous, then $\sqrt{I(\hat{\theta}_n)/I(\theta)} \xrightarrow{P_\theta} 1$, and so by Theorem 8.13 and (9.3),

$$\sqrt{nI(\hat{\theta}_n)}(\hat{\theta}_n - \theta) = \sqrt{\frac{I(\hat{\theta}_n)}{I(\theta)}}\sqrt{nI(\theta)}(\hat{\theta}_n - \theta) \Rightarrow N(0,1)$$

as $n \to \infty$. From this,

$$\begin{aligned} P_\theta\bigl(\sqrt{nI(\hat\theta_n)}|\hat\theta_n-\theta|<z_{\alpha/2}\bigr) \\ &= P_\theta\left[\theta\in\left(\hat\theta_n-\frac{z_{\alpha/2}}{\sqrt{nI(\hat\theta_n)}},\hat\theta_n+\frac{z_{\alpha/2}}{\sqrt{nI(\hat\theta_n)}}\right)\right] \\ &\to 1-\alpha \end{aligned}$$

as $n\to\infty$. So

$$\left(\hat\theta_n-\frac{z_{\alpha/2}}{\sqrt{nI(\hat\theta_n)}},\hat\theta_n+\frac{z_{\alpha/2}}{\sqrt{nI(\hat\theta_n)}}\right) \tag{9.5}$$

is a $1-\alpha$ asymptotic confidence interval for θ.

The interval (9.5) requires explicit calculation of the Fisher information. In addition, it might be argued that confidence intervals should be based solely on the shape of the likelihood function, and not on quantities that involve an expectation, such as $I(\hat\theta_n)$. Using Remark 9.16, $-l_n''(\hat\theta_n)/n \xrightarrow{P_\theta} I(\theta)$. So $\sqrt{-l_n''(\hat\theta_n)}/\sqrt{nI(\theta)} \xrightarrow{P_\theta} 1$, and multiplying (9.3) by this ratio,

$$\sqrt{-l_n''(\hat\theta_n)}(\hat\theta_n-\theta)\Rightarrow N(0,1) \tag{9.6}$$

under P_θ as $n\to\infty$. From this,

$$\left(\hat\theta_n-\frac{z_{\alpha/2}}{\sqrt{-l_n''(\hat\theta_n)}},\hat\theta_n+\frac{z_{\alpha/2}}{\sqrt{-l_n''(\hat\theta_n)}}\right) \tag{9.7}$$

is a $1-\alpha$ asymptotic confidence interval for θ. The statistic $-l_n''(\hat\theta_n)$ used to set the width of this interval is called the *observed* or *sample Fisher information.*

The interval (9.7) relies on the log-likelihood only through $\hat\theta_n$ and the curvature at $\hat\theta_n$. Our final confidence regions are called *profile regions* as they take more account of the actual shape of the likelihood function. By Taylor expansion about $\hat\theta_n$,

$$2l_n(\hat\theta_n)-2l_n(\theta)=\bigl[\sqrt{-l_n''(\theta_n^*)}(\theta-\hat\theta_n)\bigr]^2,$$

where θ_n^* is an intermediate value between θ and $\hat\theta_n$ (provided $l_n'(\hat\theta_n)=0$, which happens with probability approaching one if $\theta\in\Omega^o$). By the argument leading to (9.6),

$$\sqrt{-l_n''(\theta_n^*)}(\hat\theta_n-\theta)\Rightarrow Z\sim N(0,1),$$

and so, using Corollary 8.11,

$$2l_n(\hat\theta_n)-2l_n(\theta)\Rightarrow Z^2\sim\chi_1^2.$$

Noting that $P(Z^2<z_{\alpha/2}^2)=P(z_{\alpha/2}<Z<z_{\alpha/2})=1-\alpha$,

$$P_\theta\big(2l_n(\hat\theta_n) - 2l_n(\theta) < z^2_{\alpha/2}\big) \to 1-\alpha.$$

If we define

$$S = \big\{\theta \in \Omega : 2l_n(\hat\theta_n) - 2l_n(\theta) < z^2_{\alpha/2}\big\}, \tag{9.8}$$

then $P_\theta(\theta \in S) \to 1-\alpha$ and S is a $1-\alpha$ asymptotic confidence region for θ. Figure 9.2 illustrates how this set $S = (\delta_0, \delta_1)$ would be found from the log-likelihood function $l_n(\cdot)$.

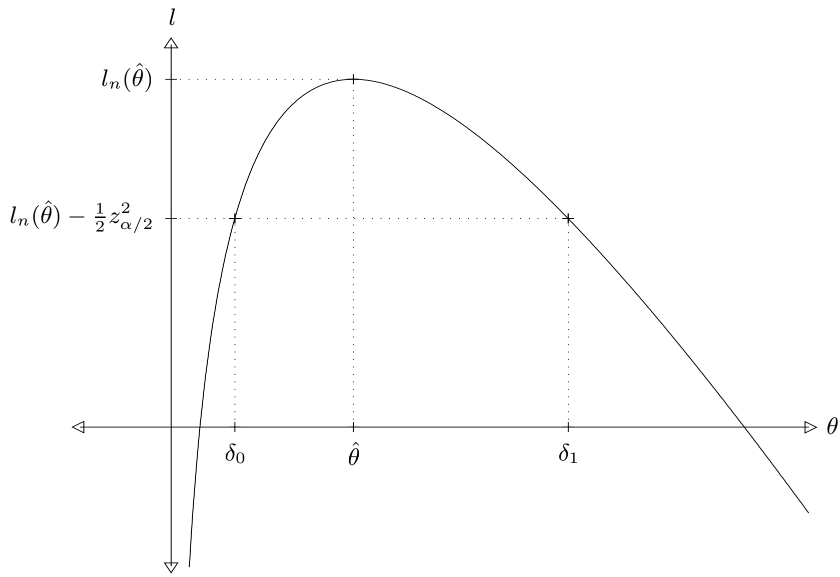

Fig. 9.2. Profile confidence interval (δ_0, δ_1).

Example 9.20. Suppose $X_1, \dots, X_n$ are i.i.d. from a Poisson distribution with mean θ. Then

$$l_n(\theta) = n\overline{X}\log\theta - n\theta - \log\left(\prod_{i=1}^n X_i!\right),$$

where $\overline{X} = (X_1 + \cdots + X_n)/n$. Since

$$l_n'(\theta) = \frac{n\overline{X}}{\theta} - n,$$

the maximum likelihood estimator of θ is $\hat\theta = \overline{X}$. Also, $I(\theta) = 1/\theta$. The first confidence region considered, (9.4), is

$$\begin{aligned} S &= \big\{\theta > 0 : \sqrt{n/\theta}|\hat\theta - \theta| < z_{\alpha/2}\big\} \\ &= \big\{\theta > 0 : \hat\theta^2 - 2\hat\theta\theta + \hat\theta^2 < z^2_{\alpha/2}\theta/n\big\} = (\hat\theta^-, \hat\theta^+), \end{aligned}$$

where

$$\hat{\theta}^{\pm} = \hat{\theta} + \frac{z_{\alpha/2}^2}{2n} \pm \sqrt{\left(\hat{\theta} + \frac{z_{\alpha/2}^2}{2n}\right)^2 - \hat{\theta}^2}.$$

The next confidence interval, (9.5), based on $I(\hat{\theta}) = 1/\overline{X}$ is

$$\left(\overline{X} - z_{\alpha/2}\sqrt{\overline{X}/n}, \overline{X} + z_{\alpha/2}\sqrt{\overline{X}/n}\right).$$

For this example, the third confidence interval is the same because the observed Fisher information, $-l_n''(\hat{\theta}) = n\overline{X}/\hat{\theta} = n/\overline{X}$, agrees with $nI(\hat{\theta})$. Note that the lower endpoint for this confidence interval will be negative if $\overline{X}$ is close enough to zero. Finally, the profile confidence interval (9.8) is

$$\left\{\theta > 0 : \theta - \overline{X}\log(\theta/\overline{X}) - \overline{X} < \frac{z_{\alpha/2}^2}{2n}\right\}.$$

This set will be an interval, because the left-hand side of the inequality is a convex function of θ, but the endpoints cannot be given explicitly and must be computed numerically.

Example 9.21. Imagine an experiment in which X is either 1 or 2, according to the toss of a fair coin, and that

$$Y|X = x \sim N(\theta, x).$$

Multiplying the marginal density (mass function) of X by the conditional density of Y given X, the joint density of X and Y is

$$f_\theta(x, y) = \frac{1}{2\sqrt{2\pi x}} \exp\left[-\frac{(y-\theta)^2}{2x}\right].$$

The Fisher information is

$$I(\theta) = -E_\theta \frac{\partial^2}{\partial\theta^2} \log f_\theta(X, Y) = E_\theta\left[\frac{1}{X}\right] = \frac{3}{4}.$$

If $(X_1, Y_1), \dots, (X_n, Y_n)$ is a random sample from this distribution, then

$$l_n(\theta) = \sum_{i=1}^n \log f_\theta(X_i, Y_i) = \sum_{i=1}^n \left[-\frac{(Y_i - \theta)^2}{2X_i} - \frac{1}{2}\log(8\pi X_i)\right]$$

and

$$l_n'(\theta) = \sum_{i=1}^n \frac{Y_i - \theta}{X_i}.$$

Equating this to zero, the maximum likelihood estimator is

$$\hat{\theta}_n = \frac{\sum_{i=1}^n (Y_i/X_i)}{\sum_{i=1}^n (1/X_i)}.$$

Also, $l_n''(\theta) = -\sum_{i=1}^n (1/X_i)$. Here the first two confidence intervals, (9.4) and (9.5), are the same (since the Fisher information is constant), namely

$$\left(\hat{\theta}_n - z_{\alpha/2}\sqrt{\frac{4}{3n}}, \hat{\theta}_n + z_{\alpha/2}\sqrt{\frac{4}{3n}}\right).$$

The last two intervals are also the same (because the log-likelihood is exactly quadratic), namely

$$\left(\hat{\theta}_n - \frac{z_{\alpha/2}}{\sqrt{\sum_{i=1}^n (1/X_i)}}, \hat{\theta}_n + \frac{z_{\alpha/2}}{\sqrt{\sum_{i=1}^n (1/X_i)}}\right). \tag{9.9}$$

In this example, the latter, likelihood-based intervals are clearly superior. Given $X_1 = x_1, \dots, X_n = x_n$, $\hat{\theta}_n$ is exactly $N\big(\theta, 1/\sum_{i=1}(1/x_i)\big)$, and by smoothing, the coverage probability for (9.9) is exactly $1-\alpha$. Also, the width of (9.9) varies in an appropriate fashion: it is shorter when many of the X_i are 1s, increasing in length when more of the X_i are 2s.

9.6 EM Algorithm: Estimation from Incomplete Data

The EM algorithm (Dempster et al. (1977)) is a recursive method to calculate maximum likelihood estimators from incomplete data. The "full data" X has density from an exponential family, but is not observed. Instead, the observed data Y is a known function of X, $Y = g(X)$, with g many-to-one (so that X cannot be recovered from Y). Here we assume for convenience the density for X is in canonical form, given by

$$h(x)e^{\eta T(x) - A(\eta)}.$$

We also assume that $\eta \in \Omega \subset \mathbb{R}$, although the algorithm works in higher dimensions, and that Y is discrete. (The full data X can be discrete or continuous.)

The EM algorithm may be useful when data are partially observed in some sense. For instance, $X_1, \dots, X_n$ could be a random sample from some exponential family, and Y_i could be X_i rounded to the nearest integer. Similar possibilities could include censored or truncated data.

The EM algorithm can also be used in situations with missing data. For instance, we may be studying answers for two multiple choice questions on some survey. The full data X gives information on answers for both questions for every subject. The incomplete data Y may provide counts for all answer combinations for respondents who answered both questions, along with counts for the first question for respondents who skipped the second question, and counts for the second question for respondents who skipped the first question.

Let $\mathcal{X}$ denote the sample space for X, $\mathcal{Y}$ the sample space for Y, and $\mathcal{X}(y)$ the cross-section

$$\mathcal{X}(y) = \{x \in \mathcal{X} : g(x) = y\}.$$

Then $Y = y$ if and only if $X \in \mathcal{X}(y)$.

Proposition 9.22. *The joint density of X and Y (with respect to $\mu \times \nu$ with μ the dominating measure for X and ν counting measure on $\mathcal{Y}$) is*

$$1_{\mathcal{X}(y)}(x)h(x)e^{\eta T(x)-A(\eta)}.$$

Proof. Let f be an arbitrary nonnegative function on $\mathcal{X} \times \mathcal{Y}$. Then $f(X,Y) = \sum_{y\in\mathcal{Y}} f(X,y)I\{Y=y\}$. Since expectation is linear (or by Fubini's theorem) and $Y = g(X)$,

$$\begin{aligned} Ef(X,Y) &= \sum_{y\in\mathcal{Y}} Ef(X,y)I\{g(X)=y\} \\ &= \sum_{y\in\mathcal{Y}} \int_{\mathcal{X}} f(x,y)I\{g(x)=y\}h(x)e^{\eta T(x)-A(\eta)}\,d\mu(x), \end{aligned}$$

and the proposition follows because $I\{g(x)=y\} = 1_{\mathcal{X}(y)}(x)$. ☐

To define the algorithm, recall that the maximum likelihood estimate of η from the full data X is $\psi(T)$, where ψ is the inverse of A'. Also, define

$$e(y,\eta) = E_\eta\big[T(X) \bigm| Y=y\big].$$

This can be computed as an integral against the conditional density of X given $Y=y$. Dividing the joint density of X and Y by the marginal density of Y, this conditional density is

$$\frac{1_{\mathcal{X}(y)}(x)h(x)e^{\eta T(x)-A(\eta)}}{f_\eta(y)},$$

where

$$f_\eta(y) = P_\eta(Y=y) = P_\eta\big(X\in\mathcal{X}(y)\big) = \int_{\mathcal{X}(y)} h(x)e^{\eta T(x)-A(\eta)}\,d\mu(x).$$

The algorithm begins with an initial guess $\hat\eta_0$ for the true maximum likelihood estimate $\hat\eta$. Using this initial guess and data Y, the value of $T(X)$ is imputed to be $T_1 = e(Y,\hat\eta_0)$ (this is called an E-step). The refined estimate for $\hat\eta$ is $\hat\eta_1 = \psi(T_1)$ (an M-step). These E- and M-steps are repeated as necessary, starting with the current estimate for $\hat{\eta}$ instead of the initial guess, until the values converge.

If the exponential family is not specified in canonical form, so the density is $h(x)e^{\eta(\theta)T(x)-B(\theta)}$, the E-step of the EM algorithm stays the same,

$$T_{k+1} = E_{\hat{\theta}_k}[T(X)|Y],$$

and for the M-step, $\hat{\theta}_k$ maximizes $\eta(\theta)T_k - B(\theta)$ over $\theta \in \Omega$.

If the EM algorithm converges to $\tilde{\eta}$, then $\tilde{\eta}$ will satisfy

$$\tilde{\eta} = \psi\big(e(Y, \tilde{\eta})\big),$$

or, equivalently,

$$A'(\tilde{\eta}) = e(Y, \tilde{\eta}).$$

Since

$$\begin{aligned}
\frac{\partial}{\partial \eta} \log f_\eta(Y) &= \frac{\frac{\partial}{\partial \eta} \int_{\mathcal{X}(Y)} h(x)e^{\eta T(x)-A(\eta)}\, d\mu(x)}{f_\eta(Y)} \\
&= \frac{\int_{\mathcal{X}(Y)} \big[T(x) - A'(\eta)\big] h(x)e^{\eta T(x)-A(\eta)}\, d\mu(x)}{f_\eta(Y)} \\
&= e(Y, \eta) - A'(\eta),
\end{aligned}$$

the log-likelihood has zero slope when $\eta = \tilde{\eta}$.

Example 9.23 (Rounding). Suppose $X_1, \dots, X_n$ are i.i.d. exponential variables with common density $f_\eta(x) = \eta e^{-\eta x}$, $x > 0$; $f_\eta(x) = 0$, $x \le 0$, and let $Y_i = \lfloor X_i \rfloor$, the greatest integer less than or equal to X_i, so we only observe the variables rounded down to the nearest integer. The joint distributions of $X_1, \dots, X_n$ form an exponential family with canonical parameter η and complete sufficient statistic $T = -(X_1 + \cdots + X_n)$. The maximum likelihood estimator of η based on X is $\psi(T) = -n/T$. Arguing as in Proposition 9.22,

$$\begin{aligned}
E_\eta[X_i|Y_i = y_i] &= E_\eta[X_i|y_i \le X_i < y_i + 1] \\
&= \frac{\int_{y_i}^{y_i+1} x\eta e^{-\eta x}\, dx}{\int_{y_i}^{y_i+1} \eta e^{-\eta x}\, dx} = y_i + \frac{e^\eta - 1 - \eta}{\eta(e^\eta - 1)},
\end{aligned}$$

and by the independence,

$$E_\eta[X_i|Y_1 = y_1, \dots, Y_n = y_n] = E_\eta[X_i|Y_i = y_i].$$

Thus

$$\begin{aligned}
e(y, \eta) &= E_\eta[T|Y_1 = y_1, \dots, Y_n = y_n] \\
&= -\sum_{i=1}^n E_\eta[X_i|Y_i = y_i] = -n\left[\overline{y} + \frac{e^\eta - 1 - \eta}{\eta(e^\eta - 1)}\right].
\end{aligned}$$

The EM algorithm is given by

$$\hat{\eta}_j = -\frac{n}{T_j} \text{ and } T_{j+1} = -n\left[\overline{Y} + \frac{e^{\hat{\eta}_j} - 1 - \hat{\eta}_j}{\hat{\eta}_j(e^{\hat{\eta}_j} - 1)}\right].$$

In this example, the mass function for Y_i can be computed explicitly:

$$P_\eta(Y_i = y) = P_\eta(y \le X_i < y+1) = (1 - e^{-\eta})(e^{-\eta})^y, \qquad y = 0, 1, \dots,$$

and we see that $Y_1, \dots, Y_n$ are i.i.d. from a geometric distribution with $p = 1 - e^{-\eta}$. The maximum likelihood estimator for p is

$$\hat{p} = \frac{1}{1 + \overline{Y}},$$

and since $\eta = -\log(1-p)$, the maximum likelihood estimator for η is

$$\hat{\eta} = -\log(1 - \hat{p}) = \log(1 + 1/\overline{Y}).$$

To study the convergence of the EM iterates $\hat{\eta}_j$, $j \ge 1$, to the maximum likelihood estimator $\hat{\eta}$, suppose $\hat{\eta}_j = \hat{\eta} + \epsilon$. By Taylor expansion,

$$\begin{aligned} T_{j+1} &= -n\left[\overline{Y} + \frac{1}{\hat{\eta} + \epsilon} - \frac{1}{e^{\hat{\eta}+\epsilon} - 1}\right] \\ &= -\frac{n}{\hat{\eta}}\left[1 - \frac{\epsilon}{\hat{\eta}} + \frac{\epsilon\hat{\eta}e^{\hat{\eta}}}{(e^{\hat{\eta}} - 1)^2} + O(\epsilon^2)\right], \end{aligned}$$

and from this,

$$\hat{\eta}_{j+1} = \hat{\eta} + \epsilon\left[1 - \frac{\hat{\eta}^2 e^{\hat{\eta}}}{(e^{\hat{\eta}} - 1)^2}\right] + O(\epsilon^2). \tag{9.10}$$

In particular, if $\hat{\eta}_j = \hat{\eta}$, so $\epsilon = 0$, $\hat{\eta}_{j+1}$ also equals $\hat{\eta}$. This shows that $\hat{\eta}$ is a fixed point of the recursion.

As a numerical routine for optimization, the EM algorithm is generally stable and reliable. One appealing property is that the likelihood increases with each successive iteration. This follows because it is in the class of MM algorithms, discussed in Lange (2004). But convergence is not guaranteed: if the likelihood has multiple modes, the algorithm may converge to a local maximum. Sufficient conditions for convergence are given by Wu (1983). Although the EM algorithm is stable, convergence can be slow. By (9.10), there is linear convergence in our example, with the convergence error $\hat{\eta}_j - \hat{\eta}$ decreasing by a constant factor (approximately) with each iteration. Linear convergence is typical for the EM algorithm. If the likelihood for Y is available, quadratic convergence, with $\hat{\eta}_{j+1} - \hat{\eta} = O\big((\hat{\eta}_j - \hat{\eta})^2\big)$, may be possible by Newton–Raphson or another search algorithm, but faster routines are generally less stable and often require information about derivatives of the objective function.

The EM algorithm can be developed without the exponential family structure assumed here. It can also be supplemented to provide numerical approximations for observed Fisher information. For these and other extensions, see McLachlan and Krishnan (2008).

9.7 Limiting Distributions in Higher Dimensions

Most of the results presented earlier in this chapter have natural extensions in higher dimensions. If $x = (x_1, \dots, x_p)$ and $y = (y_1, \dots, y_p)$ are vectors in $\mathbb{R}^p$, then $x \le y$ will mean that $x_i \le y_i$, $i = 1, \dots, p$. The cumulative distribution function H of a random vector Y in $\mathbb{R}^p$ is defined by $H(y) = P(Y \le y)$.

Definition 9.24. *Let $Y, Y_1, Y_2, \dots$ be random vectors taking values in $\mathbb{R}^p$, with H the cumulative distribution of Y and H_n the cumulative distribution function of Y_n, $n = 1, 2, \dots$. Then Y_n* converges in distribution *to Y as $n \to \infty$, written $Y_n \Rightarrow Y$, if $H_n(y) \to H(y)$ as $n \to \infty$ at any continuity point y of H.*

For a set $S \subset \mathbb{R}^p$, let $\partial S = \overline{S} - S^o$ denote the boundary of S. The following result lists conditions equivalent to convergence in distribution.

Theorem 9.25. *If $Y, Y_1, Y_2, \dots$ are random vectors in $\mathbb{R}^p$, then the following conditions are equivalent.*

1. $Y_n \Rightarrow Y$ *as* $n \to \infty$.
2. $Eu(Y_n) \to Eu(Y)$ *for every bounded continuous function* $u : \mathbb{R}^p \to \mathbb{R}$.
3. $\liminf_{n\to\infty} P(Y_n \in G) \ge P(Y \in G)$ *for every open set* G.
4. $\limsup_{n\to\infty} P(Y_n \in F) \le P(Y \in F)$ *for every closed set* F.
5. $P(Y_n \in S) \to P(Y \in S)$ *for any Borel set* S *such that* $P(Y \in \partial S) = 0$.

This result is called the *portmanteau theorem*. The second condition in this result is often taken as the definition of convergence in distribution. As is the case for one dimension, the following result is an easy corollary.

Corollary 9.26. *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a continuous function, and if $Y_n \Rightarrow Y$ (a random vector in $\mathbb{R}^p$), then*

$$f(Y_n) \Rightarrow f(Y).$$

In the multivariate extension of the central limit theorem, averages of i.i.d. random vectors, after suitable centering and scaling, will converge to a limit, called the multivariate normal distribution. One way to describe this distribution uses moment generating functions. The moment generating function M_Y for a random vector Y in $\mathbb{R}^p$ is given by

$$M_Y(u) = Ee^{u'Y}, \qquad u \in \mathbb{R}^p.$$

As in the univariate case, if the moment generating functions of two random vectors X and Y agree on any nonempty open set, then X and Y have the same distribution. Suppose $Z = (Z_1, \dots, Z_p)'$ with $Z_1, \dots, Z_p$ a random sample from $N(0, 1)$. By independence,

$$Ee^{u'Z} = (Ee^{u_1 Z_1}) \times \cdots \times (Ee^{u_p Z_p}) = e^{u_1^2/2} \times \cdots \times e^{u_p^2/2} = e^{u'u/2}.$$

Suppose we define $X = \mu + AZ$. Then

$$EX = \mu \text{ and } \operatorname{Cov}(X) = \Sigma = AA'.$$

Taking $u = A't$ in the formula above for $Ee^{u'Z}$, X has moment generating function

$$\begin{aligned} M_X(t) &= Ee^{t'X} = Ee^{t'\mu + t'AZ} = e^{t'\mu}Ee^{u'Z} \\ &= e^{t'\mu + u'u/2} = e^{t'\mu + t'AA't/2} = e^{t'\mu + t'\Sigma t/2}. \end{aligned}$$

Note that this function depends on A only through the covariance $\Sigma = AA'$. The distribution for X is called the *multivariate normal distribution* with mean μ and covariance matrix Σ, written

$$X \sim N(\mu, \Sigma).$$

Linear transformations preserve normality. If $X \sim N(\mu, \Sigma)$ and $Y = AX + b$, then

$$\begin{aligned} M_Y(u) &= Ee^{u'(AX+b)} = e^{u'b}Ee^{u'AX} \\ &= e^{u'b}M_X(A'u) = \exp[u'b + u'A\mu + u'A\Sigma A'u/2], \end{aligned}$$

and so

$$Y \sim N(b + A\mu, A\Sigma A').$$

Naturally, the parameters for this distribution are the mean and covariance of Y.

In the construction for $N(\mu, \Sigma)$, any nonnegative definite matrix Σ is possible. One suitable matrix A would be a symmetric square root of Σ. This can be found writing $\Sigma = ODO'$ with O an orthogonal matrix (so $O'O = I$) and D diagonal, and defining $\Sigma^{1/2} = OD^{1/2}O'$, where $D^{1/2}$ is diagonal with entries the square roots of the diagonal entries of D. Then $\Sigma^{1/2}$ is symmetric and

$$\Sigma^{1/2}\Sigma^{1/2} = OD^{1/2}O'OD^{1/2}O' = OD^{1/2}D^{1/2}O' = ODO' = \Sigma. \tag{9.11}$$

As a side note, the construction here can be used to define other powers, including negative powers, of a symmetric positive definite matrix Σ. In this case, the diagonal entries D_{ii} of D are all positive, D^α can be taken as the diagonal matrix with diagonal entries D_{ii}^α, and $\Sigma^\alpha \overset{\text{def}}{=} OD^\alpha O'$. This construction gives $\Sigma^0 = I$, and the powers of Σ satisfy

$$\Sigma^\alpha \Sigma^\beta = \Sigma^{\alpha+\beta}.$$

When Σ is positive definite ($\Sigma > 0$), $N(\mu, \Sigma)$ is absolutely continuous. To derive the density, note that the density of Z is

$$\prod_{i=1}^{p} \frac{e^{-z_i^2/2}}{\sqrt{2\pi}} = \frac{e^{-z'z/2}}{(2\pi)^{p/2}}.$$

Also, the linear transformation $z \rightsquigarrow \mu + \Sigma^{1/2} z$ is one-to-one with inverse $x \rightsquigarrow \Sigma^{-1/2}(x-\mu)$. (Here $\Sigma^{-1/2}$ is the inverse of $\Sigma^{1/2}$.) The Jacobian of the inverse transformation is $\det(\Sigma^{-1/2}) = 1/\sqrt{\det \Sigma}$. So if $X = \mu + \Sigma^{1/2} Z \sim N(\mu, \Sigma)$, a multivariate change of variables gives

$$\begin{aligned} P(X \in B) = P(\mu + \Sigma^{1/2} Z \in B) &= \int \cdots \int 1_B(\mu + \Sigma^{1/2} z) \frac{e^{-z'z/2}}{(2\pi)^{p/2}}\, dz \\ &= \int \cdots \int 1_B(x) \frac{\exp\Bigl(-\frac{1}{2}\bigl(\Sigma^{-1/2}(x-\mu)\bigr)'\bigl(\Sigma^{-1/2}(x-\mu)\bigr)\Bigr)}{(2\pi)^{p/2}\sqrt{\det \Sigma}}\, dx. \end{aligned}$$

From this, $X \sim N(\mu, \Sigma)$ has density

$$\frac{\exp\Bigl(-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\Bigr)}{(2\pi)^{p/2}\sqrt{\det \Sigma}}.$$

The following result generalizes the central limit theorem (Theorem 8.12) to higher dimensions. For a proof, see Billingsley (1995) or any standard text on probability.

Theorem 9.27 (Multivariate Central Limit Theorem). *If $X_1, X_2, \dots$ are i.i.d. random vectors with common mean μ and common covariance matrix Σ, and if $\overline{X}_n = (X_1 + \cdots + X_n)/n$, $n \ge 1$, then*

$$\sqrt{n}(\overline{X} - \mu) \Rightarrow Y \sim N(0, \Sigma).$$

Asymptotic normality of the maximum likelihood estimator will involve random matrices. The most convenient way to deal with convergence in probability of random matrices is to treat them as vectors, introducing the Euclidean (or Frobenius) norm

$$\|M\| = \left\{ \sum_{i,j} M_{ij}^2 \right\}^{1/2}.$$

Definition 9.28. *A sequence of random matrices M_n, $n \ge 1$ converges in probability to a random matrix M, written $M_n \xrightarrow{p} M$, if for every $\epsilon > 0$,*

$$P\bigl(\|M_n - M\| > \epsilon\bigr) \to 0$$

as $n \to \infty$. Equivalently, $M_n \xrightarrow{p} M$ as $n \to \infty$ if $[M_n]_{ij} \xrightarrow{p} M_{ij}$ as $n \to \infty$, for all i and j.

The following results are natural extensions of the corresponding results in one dimension.

Theorem 9.29. *If $M_n \xrightarrow{p} M$ as $n \to \infty$, with M a constant matrix, and if f is continuous at M, then $f(M_n) \xrightarrow{p} f(M)$.*

Theorem 9.30. *If $Y, Y_1, Y_2, \dots$ are random vectors in $\mathbb{R}^p$ with $Y_n \Rightarrow Y$ as $n \to \infty$, and if M_n are random matrices with $M_n \xrightarrow{p} M$ as $n \to \infty$, with M a constant matrix, then*

$$M_n Y_n \Rightarrow MY$$

as $n \to \infty$

Technical details establishing asymptotic normality of the maximum likelihood estimator in higher dimensions are essentially the same as the details in one dimension, so the presentation here just highlights the main ideas in an informal fashion. Let $X, X_1, X_2, \dots$ be i.i.d. with common density f_θ, $\theta \in \Omega \subset \mathbb{R}^p$. As in one dimension, the log-likelihood can be written as a sum,

$$l_n(\theta) = \sum_{i=1}^n \log f_\theta(X_i).$$

As in Section 4.6, the Fisher information is a matrix,

$$I(\theta) = \operatorname{Cov}_\theta\bigl(\nabla_\theta \log f_\theta(X)\bigr) = -E_\theta \nabla_\theta^2 \log f_\theta(X),$$

and

$$E_\theta \nabla_\theta \log f_\theta(X) = 0.$$

The maximum likelihood estimator based on $X_1, \dots, X_n$ maximizes l_n. If $\hat\theta_n$ is consistent and θ lies in the interior of Ω, then with probability tending to one,

$$\nabla_\theta l_n(\hat\theta_n) = 0.$$

Taylor expansion of $\nabla_\theta l_n(\cdot)$ about θ gives the following approximation:

$$\nabla_\theta l_n(\hat\theta_n) \approx \nabla_\theta l_n(\theta) + \nabla_\theta^2 l_n(\theta)(\hat\theta_n - \theta).$$

Setting this expression to zero, solving, and introducing powers of n,

$$\sqrt{n}(\hat\theta_n - \theta) \approx \left[-\frac{1}{n}\nabla_\theta^2 l_n(\theta)\right]^{-1} \frac{1}{\sqrt{n}}\nabla_\theta l_n(\theta). \tag{9.12}$$

By the multivariate central limit theorem,

$$\frac{1}{\sqrt{n}}\nabla_\theta l_n(\theta) = \sqrt{n}\left[\frac{1}{n}\sum_{i=1}^n \nabla_\theta \log f_\theta(X_i) - 0\right] \Rightarrow Y \sim N\bigl(0, I(\theta)\bigr)$$

as $n \to \infty$. Also, by the law of large numbers,

$$-\frac{1}{n}\nabla_\theta^2 l_n(\theta) = \frac{1}{n}\sum_{i=1}^n \bigl[-\nabla_\theta^2 \log f_\theta(X_i)\bigr] \xrightarrow{P_\theta} I(\theta)$$

as $n \to \infty$. Since the function $A \rightsquigarrow A^{-1}$ is continuous for nonsingular matrices A, if $I(\theta) > 0$,

$$\left[-\frac{1}{n}\nabla_\theta^2 l_n(\theta)\right]^{-1} \xrightarrow{P_\theta} I(\theta)^{-1}.$$

The error in (9.12) tends to zero in probability, and then using Theorem 9.30,

$$\sqrt{n}(\hat\theta_n - \theta) \Rightarrow I(\theta)^{-1}Y \sim N\bigl(0, I(\theta)^{-1}\bigr).$$

To verify the stated distribution for $I(\theta)^{-1}Y$, note that Y has the same distribution as $I(\theta)^{1/2}Z$ with Z a vector of i.i.d. standard normal variates, and so

$$I(\theta)^{-1}Y \sim I(\theta)^{-1}I(\theta)^{1/2}Z = I(\theta)^{-1/2}Z \sim N\bigl(0, I(\theta)^{-1}\bigr).$$

The following proposition is a multivariate extension of the delta method.

Proposition 9.31. *If $g : \Omega \to \mathbb{R}$ is differentiable at θ, $I(\theta)$ is positive definite, and $\sqrt{n}(\hat\theta_n - \theta) \Rightarrow N\bigl(0, I(\theta)^{-1}\bigr)$, then*

$$\sqrt{n}\bigl(g(\hat\theta_n) - g(\theta)\bigr) \Rightarrow N\bigl(0, \nu^2(\theta)\bigr)$$

with

$$\nu^2(\theta) = \bigl(\nabla g(\theta)\bigr)' I(\theta)^{-1}\nabla g(\theta).$$

As an application of this result, if $\hat\nu_n$ is a consistent estimator of $\nu(\theta)$ and $\nu(\theta) > 0$, then

$$\left(g(\hat\theta_n) - \frac{z_{\alpha/2}\hat\nu_n}{\sqrt{n}},\ g(\hat\theta_n) + \frac{z_{\alpha/2}\hat\nu_n}{\sqrt{n}}\right) \tag{9.13}$$

is a $1-\alpha$ asymptotic confidence interval for $g(\theta)$.

Finally, the delta method can be extended to vector-valued functions. In this result, $Dg(\theta)$ denotes a matrix of partial derivatives of g, with entries $[Dg(\theta)]_{ij} = \partial g_i(\theta)/\partial\theta_j$.

Proposition 9.32. *If $g : \Omega \to \mathbb{R}^m$ is differentiable at θ, $I(\theta)$ is positive definite, and $\sqrt{n}(\hat\theta_n - \theta) \Rightarrow N\bigl(0, I(\theta)^{-1}\bigr)$, then*

$$\sqrt{n}\bigl(g(\hat\theta_n) - g(\theta)\bigr) \Rightarrow N\bigl(0, \Sigma(\theta)\bigr)$$

with

$$\Sigma(\theta) = Dg(\theta)I(\theta)^{-1}[Dg(\theta)]'.$$

9.8 M-Estimators for a Location Parameter

Let $X, X_1, X_2, \ldots$ be i.i.d. from some distribution Q, and let ρ be a convex function[4] on $\mathbb{R}$ with $\rho(x) \to \infty$ as $x \to \pm\infty$. The *M-estimator* T_n associated

[4] These conditions on ρ are convenient, because with them H must have a minimum. But they could be relaxed.

with ρ minimizes

$$H(t) = \sum_{i=1}^{n} \rho(X_i - t)$$

over $t \in \mathbb{R}$. If ρ is continuously differentiable and $\psi = \rho'$, then T_n is also a root of the estimating equation

$$\overline{W}_n(t) \overset{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} \psi(X_i - t) = 0.$$

Several common estimates of location are M-estimators. If $\rho(x) = x^2$, then $T_n = \overline{X}_n$, the sample average, and if $\rho(x) = |x|$, then T_n is the median. Finally, if Q lies in a location family of absolutely continuous distributions with log-concave densities $f(x - \theta)$, then taking $\rho = -\log f$, T_n is the maximum likelihood estimator of θ.

To study convergence, let us assume ρ is continuously differentiable, and define

$$\lambda(t) = E\psi(X - t) = \int \psi(x - t)\, dQ(x).$$

Since ρ is convex, ψ is nondecreasing and λ is nonincreasing. Also, $\lambda(t)$ will be negative for t sufficiently large and positive for t sufficiently small.

Lemma 9.33. *If $\lambda(t)$ is finite for all $t \in \mathbb{R}$ and $\lambda(t) = 0$ has a unique root c, then $T_n \xrightarrow{p} c$.*

Using part 3 of Theorem 9.4, this lemma follows fairly easily from our law of large numbers for random functions, Theorem 9.2. The monotonicity of ψ can be used both to restrict attention to a compact set K and to argue that the envelope of the summands over K is integrable.

If ρ is symmetric, $\rho(x) = \rho(-x)$ for all $x \in \mathbb{R}$, and if the distribution of X is symmetric about some value θ, so that $X - \theta \sim \theta - X$, then in this lemma the limiting value c is θ.

Asymptotic normality for T_n can be established with an argument similar to that used to show asymptotic normality for the maximum likelihood estimator. If ψ is continuously differentiable, then Taylor expansion of $\overline{W}_n$ about c gives

$$\overline{W}_n(T_n) = \overline{W}_n(c) + (T_n - c)\overline{W}_n'(t_n^*),$$

with t_n^* an intermediate value between c and T_n. Since $\overline{W}_n(T_n)$ is zero,

$$\sqrt{n}(T_n - c) = -\frac{\sqrt{n}\,\overline{W}_n(c)}{\overline{W}_n'(t_n^*)}.$$

By the central limit theorem,

$$\sqrt{n}\,\overline{W}_n(c) \Rightarrow N\bigl(0, \mathrm{Var}[\psi(X - c)]\bigr),$$

and since $t_n^* \xrightarrow{p} c$, under suitable regularity[5]

$$-\overline{W}_n'(t_n^*) \xrightarrow{p} -E\psi'(X-c) = \lambda'(c).$$

Thus

$$\sqrt{n}(T_n - c) \Rightarrow N\big(0, V(\psi, Q)\big),$$

where

$$V(\psi, Q) = \frac{E\psi^2(X-c)}{[\lambda'(c)]^2}.$$

M-estimation was introduced by Huber (1964) to consider *robust* estimation of a location parameter. As noted above, if ρ is symmetric, $\rho(x) = \rho(-x)$ for all $x \in \mathbb{R}$, and if the distribution of X is symmetric about θ, $X - \theta \sim \theta - X$, then T_n is a consistent estimator of θ. For instance, we might have $Q = N(\theta, 1)$, so that $X - \theta \sim N(0,1)$. Taking ρ the square function, $\rho(x) = x^2$, T_n is the sample average $\overline{X}_n$, which is consistent and fully efficient. In a situation like this it may seem foolish to base M-estimation on any other function ρ, an impression that seems entirely reasonable if we have complete confidence in a normal model for the data. Unfortunately, doubts arise if we entertain the possibility that our normal distribution for X is even slightly "contaminated" by some other distribution. Perhaps

$$X \sim (1-\epsilon)N(\theta, 1) + \epsilon Q^*, \tag{9.14}$$

with Q^* some other distribution symmetric about θ. Then

$$\mathrm{Var}(X) = 1 - \epsilon + \epsilon \int (x-\theta)^2 \, dQ^*(x).$$

By the central limit theorem, the asymptotic performance of $\overline{X}_n$ is driven by the variance of the summands, and even a small amount of contamination ϵ can significantly degrade the performance of $\overline{X}_n$ if the variance of Q^* is large. If Q^* has infinite variance, $\sqrt{n}(\overline{X}_n - \theta)$ will not even converge in distribution.

Let $\mathcal{C} = \mathcal{C}_\epsilon$ be the class of all distributions for X with the form in (9.14). If one is confident that $Q \in \mathcal{C}_\epsilon$ it may be natural to use an M-estimator with

$$\sup_{Q \in \mathcal{C}_\epsilon} V(\psi, Q)$$

as small as possible. The following result shows that this is possible and describes the optimal function ψ_0. The optimal function ψ_0 and $\rho_0 = \psi_0'$ are plotted in Figure 9.3.

Theorem 9.34 (Huber). *The asymptotic variance $V(\psi, Q)$ has a saddle point: There exists $Q_0 = (1-\epsilon)N(\theta, 1) + \epsilon Q_0^* \in \mathcal{C}_\epsilon$ and ψ_0 such that*

[5] The condition $E \sup_{t \in [c-\epsilon, c+\epsilon]} |\psi'(X-t)| < \infty$ for some $\epsilon > 0$ is sufficient.

$$\sup_{Q\in\mathcal{C}_\epsilon} V(\psi_0,Q)=V(\psi_0,Q_0)=\inf_{\psi} V(\psi,Q_0).$$

If k solves

$$\frac{1}{1-\epsilon}=P\bigl(|Z|<k\bigr)+\frac{2\phi(k)}{k},$$

where $Z\sim N(0,1)$, and if

$$\rho_0(t)=\begin{cases}\frac{1}{2}t^2, & |t|\le k;\\ k|t|-\frac{1}{2}k^2, & |t|\ge k,\end{cases}$$

then $\psi_0=\rho_0'$ and Q_0^* is any distribution symmetric about θ with

$$Q_0^*\bigl([\theta-k,\theta+k]\bigr)=0.$$

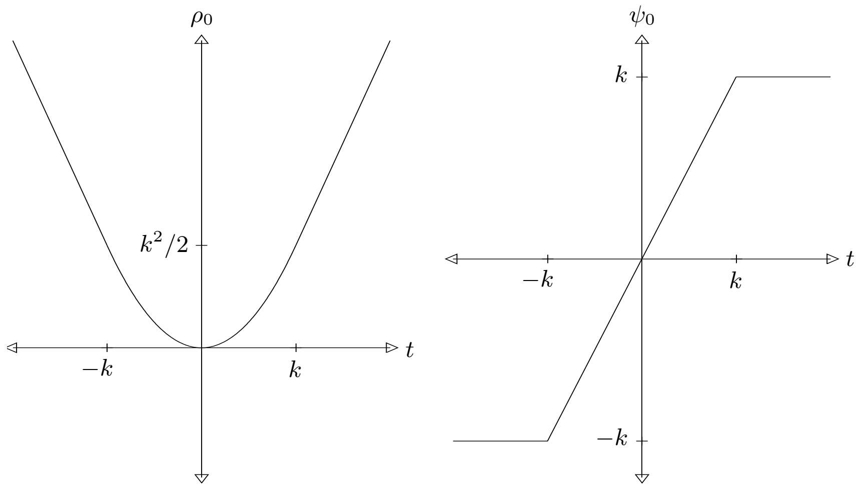

Fig. 9.3. Functions ρ_0 and ψ_0.

9.9 Models with Dependent Observations[6]

The asymptotic theory developed earlier in this chapter is based on models with i.i.d. observations. Extensions in which the observations need not have the same distribution and may exhibit dependence are crucial in various applications, and there is a huge literature extending basic results in various directions. In our discussion of the i.i.d. case, the law of large numbers and

[6] Results in this section are somewhat technical and are not used in later chapters.

the central limit theorem were our main tools from probability. Extensions typically rely on more general versions of these results, but the overall nature of the argument is similar to that for the i.i.d. case in most other ways. As a single example, this section sketches how large-sample theory can be developed for models for stationary time series. For extensions in a variety of other directions, see DasGupta (2008).

Time series analysis concerns inference for observations observed over time. Dependence is common and is allowed in the models considered here, but we restrict attention to observations that are stationary according to the following definition. A sequence of random variables, X_n, $n \in \mathbb{Z}$, will be called a (stochastic) *process*, and can be viewed as an infinite-dimensional random vector taking values in $\mathbb{R}^{\mathbb{Z}}$.

Definition 9.35. *The process X is (strictly)* stationary *if*

$$(X_1, \dots, X_k) \sim (X_{n+1}, \dots, X_{n+k}),$$

for all $k \geq 1$ and $n \in \mathbb{Z}$.

Taking $k = 1$ in this definition, observations X_i from a stationary process are identically distributed, and it feels natural to expect information to accumulate fairly linearly over time, as it would with i.i.d. data.

Viewing a sequence x_n, $n \in \mathbb{Z}$, as a single point $x \in \mathbb{R}^{\mathbb{Z}}$, we can define a shift operator T that acts on x by incrementing time. Specifically, $y = T(x)$ if $y_n = x_{n+1}$ for all $n \in \mathbb{Z}$. Using T, a process X is stationary if $X \sim T(X)$, where $X \sim Y$ means that the finite-dimensional distributions for X and Y agree:

$$(X_i, X_{i+1}, \dots X_j) \sim (Y_i, Y_{i+1}, \dots Y_j)$$

for all $i \leq j$ in $\mathbb{Z}$.

Example 9.36. If X_n, $n \in \mathbb{Z}$, are i.i.d. from some distribution Q, then X is stationary. More generally, a mixture model in which, given $Y = y$, the variables X_n, $n \in \mathbb{Z}$, are i.i.d. from Q_y also gives a stationary process X.

If ϵ_n, $n \in \mathbb{Z}$, are i.i.d. from some distribution Q with $E\epsilon_n = 0$ and $E\epsilon_n^2 < \infty$, and if c_n, $n \geq 1$, are square summable constants, then

$$X_n = \sum_{j=0}^{\infty} c_j \epsilon_{n-j}, \qquad n \in \mathbb{Z},$$

defines a stationary process X, called a *linear process.* If $c_n = \rho^n$ with $|\rho| < 1$ then

$$X_{n+1} = \rho X_n + \epsilon_{n+1},$$

and if $Q = N(0, \sigma^2)$ we have the autoregressive model introduced in Example 6.4.

The ergodic theorem is a generalization of the law of large numbers useful in this setting. To describe this result, let $\mathcal{B}_{\mathbb{Z}}$ denote the Borel sets of $\mathbb{R}^{\mathbb{Z}}$.[7] A set $B \in \mathcal{B}_{\mathbb{Z}}$ is called *shift invariant* if $x \in B$ if and only if $T(x) \in B$. Changing the value for x_n at any fixed n will not change whether x lies in a shift invariant set; inclusion can only depend on how the sequence behaves as $|n| \to \infty$. For instance, sets

$$\Bigl\{x : \limsup_{n\to-\infty} x_n \le c\Bigr\} \text{ and } \left\{x : \frac{1}{n}\sum_{i=1}^n x_i \to c \text{ as } n \to \infty\right\}$$

are shift invariant, but $\{x : x_3 + x_7 \le x_4\}$ is not.

Definition 9.37. *A stationary process X_n, $n \in \mathbb{Z}$, is* ergodic *if*

$$P(X \in B) = 0 \text{ or } 1$$

whenever B is a shift invariant set in $\mathcal{B}_{\mathbb{Z}}$.

In Example 9.36, i.i.d. variables and linear processes can be shown to be ergodic. But i.i.d. mixtures generally are not; see Problem 9.38.

With this definition we can now state the ergodic theorem. Let T_j denote T composed with itself j times.

Theorem 9.38 (Ergodic Theorem). *If X is a stationary ergodic process and $E|g(X)| < \infty$, then*

$$\frac{1}{n}\sum_{j=1}^n g\bigl(T_j(X)\bigr) \to \mu_g \stackrel{\text{def}}{=} Eg(X)$$

almost surely as $n \to \infty$. The convergence here also holds in mean,

$$E\left|\frac{1}{n}\sum_{j=1}^n g\bigl(T_j(X)\bigr) - \mu_g\right| \to 0.$$

As noted, if the X_n are i.i.d. from some distribution Q, then X is ergodic. If g is defined by $g(x) = x_0$, then $\mu_g = EX_n$, $g\bigl(T_j(X)\bigr) = X_j$, and the ergodic theorem gives the strong law of large numbers.

For convergence in distribution we use an extension of the ordinary central limit theorem to martingales.

Definition 9.39. *For $n \ge 1$, let M_n be a function of $X_1, \dots, X_n$. The process M_n, $n \ge 1$, is a (zero mean) martingale with respect to X_n, $n \ge 1$, if $EM_1 = 0$ and*

$$E[M_{n+1}|X_1, \dots, X_n] = M_n, \qquad n \ge 1.$$

[7] Formally, $\mathcal{B}_{\mathbb{Z}}$ is the smallest σ-field that contains all (finite) rectangles of form $\{x \in \mathbb{R}^{\mathbb{Z}} : x_k \in (a_k, b_k), i \le k \le j\}$.

If M_n, $n \ge 1$, is a martingale, then by smoothing

$$\begin{aligned} E[M_{n+2}|X_1,\dots,X_n] &= E\big[E[M_{n+2}|X_1,\dots,X_{n+1}] \bigm| X_1,\dots,X_n\big] \\ &= E[M_{n+1}|X_1,\dots,X_n] = M_n. \end{aligned}$$

With further iteration it is easy to see that

$$E[M_{n+k}|X_1,\dots,X_n] = M_n, \qquad n \ge 1, \quad k \ge 1. \tag{9.15}$$

Defining differences $Y_{n+1} = M_{n+1} - M_n$, $n \ge 1$, with $Y_1 = M_1$,

$$M_n = \sum_{i=1}^n Y_i, \qquad n \ge 1.$$

Using (9.15),

$$\begin{aligned} E[Y_{n+k}|X_1,\dots,X_n] &= E[M_{n+k+1}|X_1,\dots,X_n] - E[M_{n+k}|X_1,\dots,X_n] \\ &= M_n - M_n = 0, \qquad n \ge 1,\ k \ge 1. \end{aligned}$$

If the X_i are i.i.d. with mean μ and $Y_i = X_i - \mu$, it is easy to check that $M_n = Y_1 + \cdots + Y_n$, $n \ge 1$, is a martingale. By the ordinary central limit theorem, $M_n/\sqrt{n}$ is approximately normal. In the more general case, the summands Y_i may be dependent. But by smoothing,

$$EY_{n+k}Y_n = EE[Y_{n+k}Y_n|X_1,\dots,X_n] = E\big[Y_n E[Y_{n+k}|X_1,\dots,X_n]\big] = 0,$$

and so they remain uncorrelated, as in the i.i.d. case. Let $\sigma_n^2 \overset{\text{def}}{=} \mathrm{Var}(Y_n) = EY_n^2$, and note that since the summands are uncorrelated,

$$\mathrm{Var}(M_n) = \sigma_1^2 + \cdots + \sigma_n^2.$$

For convenience, we assume that $\sigma_n^2 \to \sigma^2$ as $n \to \infty$. (For more general results, see Hall and Heyde (1980).) Then

$$\mathrm{Var}\big(M_n/\sqrt{n}\big) = \frac{1}{n}\sum_{i=1}^n \sigma_i^2 \to \sigma^2.$$

In contrast with the ordinary central limit theorem, the result for martingales requires some control of the conditional variances

$$s_n^2 \overset{\text{def}}{=} \mathrm{Var}(Y_n|X_1,\dots,X_{n-1}) = E[Y_n^2|X_1,\dots,X_{n-1}].$$

Specifically, the following result from Brown (1971a) assumes that

$$\frac{1}{n}\sum_{1=1}^n s_i^2 \xrightarrow{p} \sigma^2, \tag{9.16}$$

and that

$$\frac{1}{n}\sum_{i=1}^{n} E\bigl[Y_i^2 I\{|Y_i \,|\geq \epsilon\sqrt{n}\}\bigr] \to 0 \tag{9.17}$$

as $n \to \infty$ for all $\epsilon > 0$. Requirement (9.17) is called the *Lindeberg condition.* Since $Es_i^2 = \sigma_i^2$, (9.16) might be considered a law of large numbers.

Theorem 9.40. *If M_n, $n \geq 1$, is a mean zero martingale satisfying* (9.16) *and* (9.17)*, then*

$$\frac{M_n}{\sqrt{n}} \Rightarrow N(0, \sigma^2).$$

Turning now to inference, let $\theta \in \Omega \subset \mathbb{R}$ be an unknown parameter, and let P_θ be the distribution for a process X that is stationary and ergodic for all $\theta \in \Omega$. Also, assume that finite-dimensional joint distributions for X are dominated, and let $f_\theta(x_1, \dots, x_n)$ denote the density of $X_1, \dots, X_n$ under P_θ. As usual, this density can be factored using conditional densities as

$$f_\theta(x_1, \dots, x_n) = \prod_{i=1}^{n} f_\theta(x_i | x_1, \dots, x_{i-1}).$$

The log-likelihood function is then

$$l_n(\omega) = \sum_{i=1}^{n} \log f_\omega(X_i | X_1, \dots, X_{i-1}),$$

where, as before, we let ω denote a generic value for the unknown parameter, reserving θ for the true value.

With dependent observations, the conditional distributions for X_n change as we condition on past observations. But for most models of interest the amount of change decrease as we condition further into the past, with these distributions converging to the conditional distribution given the entire history of the process. Specifically, we assume that the conditional densities

$$f_\omega(X_n | X_{n-1}, \dots, X_{n-m}) \to f_\omega(X_n | X_{n-1}, \dots) \tag{9.18}$$

in an appropriate sense as $m \to \infty$. The autoregressive model, for instance, has Markov structure with the conditional distributions for X_n depending only on the previous observation, $f_\omega(X_n | X_{n-1}, \dots, X_{n-m}) = f_\omega(X_n | X_{n-1})$. So in this case (9.18) is immediate.

In Section 9.2, our first step towards understanding consistency of the maximum likelihood estimator $\hat{\theta}_n$ was to argue that if $\omega \neq \theta$, $l_n(\theta)$ will exceed $l_n(\omega)$ with probability tending to one as $n \to \infty$. To understand why that will happen in this case, define

$$g(X) = \log \frac{f_\theta(X_0 | X_{-1}, X_{-2}, \dots)}{f_\omega(X_0 | X_{-1}, X_{-2}, \dots)}.$$

Note that

$$E_\theta\bigl[g(X) \bigm| X_{-1}, X_{-2}, \dots\bigr]$$

is the Kullback–Leibler information between θ and ω for the conditional distributions of X_0 given the past, and is positive unless these conditional distributions coincide. Assuming this is not almost surely the case,

$$\mu_g = E_\theta g(X) = E_\theta E_\theta[g(X)|X_{-1}, X_{-2}, \dots]$$

is positive. Using (9.18), if j is large,

$$\log \frac{f_\theta(X_j|X_1, \dots, X_{j-1})}{f_\omega(X_j|X_1, \dots, X_{j-1})} \approx \log \frac{f_\theta(X_j|X_{j-1}, X_{j-2}, \dots)}{f_\omega(X_j|X_{j-1}, X_{j-2}, \dots)} = g\bigl(T_j(X)\bigr).$$

Using this approximation,

$$\frac{l_n(\theta) - l_n(\omega)}{n} = \frac{1}{n}\sum_{j=1}^n \log \frac{f_\theta(X_j|X_1, \dots, X_{j-1})}{f_\omega(X_j|X_1, \dots, X_{j-1})} \approx \frac{1}{n}\sum_{i=1}^n g\bigl(T_j(X)\bigr), \quad (9.19)$$

converging to $\mu_g > 0$ as $n \to \infty$. If the approximation error here tends to zero in probability (see Problem 9.40 for a sufficient condition), then the likelihood at θ will be greater than the likelihood at ω with probability tending to one. Building on this basic idea, consistency of $\hat\theta_n$ can be established in regular cases using the same arguments as those for the i.i.d. case, changing likelihood at a point ω to the supremum of the likelihood in a neighborhood of ω (or a neighborhood of infinity).

In the univariate i.i.d. case, asymptotic normality followed using Taylor approximation to show that

$$\sqrt{n}(\hat\theta_n - \theta) = \frac{\frac{1}{\sqrt{n}} l_n'(\theta)}{-\frac{1}{n} l_n''(\theta)} + o_p(1) \quad (9.20)$$

with

$$\frac{1}{\sqrt{n}} l_n'(\theta) \Rightarrow N\bigl(0, I(\theta)\bigr) \text{ and } -\frac{1}{n} l_n''(\theta) \xrightarrow{p} I(\theta). \quad (9.21)$$

The same Taylor expansion argument can be used in this setting, so we mainly need to understand why the limits in (9.21) hold. Convergence for $-l_n''(\theta)/n$ is similar to the argument for consistency above. If we define h as

$$h(X) = -\frac{\partial^2}{\partial\theta^2} \log f_\theta(X_0|X_{-1}, X_{-2}, \dots),$$

and assume for large j,

$$\begin{aligned} -\frac{\partial^2}{\partial\theta^2} \log f_\theta(X_j|X_1, \dots, X_{j-1}) &\approx -\frac{\partial^2}{\partial\theta^2} \log f_\theta(X_j|X_{j-1}, X_{j-2}, \dots) \\ &= h\bigl(T_j(X)\bigr), \end{aligned}$$

which is essentially that (9.18) holds in a differentiable sense, then

$$-\frac{1}{n}l_n''(\theta) = -\frac{1}{n}\sum_{j=1}^n \frac{\partial^2}{\partial\theta^2}\log f_\theta(X_j|X_1,\dots,X_{j-1}) \approx \frac{1}{n}\sum_{j=1}^n h\bigl(T_j(X)\bigr),$$

converging to $E_\theta h(X)$ by the ergodic theorem. Since Fisher information here for all n observations is $I_n(\theta) = -E_\theta l_n''(\theta)$, if the approximation error tends to zero in probability, then

$$\frac{1}{n}I_n(\theta) \xrightarrow{p} E_\theta h(X). \tag{9.22}$$

So it is natural to define $I(\theta) = E_\theta h(X)$, interpreted with large samples as average Fisher information per observation.

Asymptotic normality for the score function $l_n'(\theta)$ is based on the martingale central limit theorem. Define

$$Y_j = \frac{\partial}{\partial\theta}\log f_\theta(X_j|X_1,\dots,X_{j-1})$$

so that

$$l_n'(\theta) = \sum_{j=1}^n Y_j.$$

The martingale structure needed will hold if we can pass derivatives inside integrals, as in the Cramér–Rao bound, but now with conditional densities. Specifically, we want

$$\begin{aligned} 0 &= \int \frac{\partial}{\partial\theta}f_\theta(x_j|x_1,\dots,x_{j-1})\,d\mu(x_j) \\ &= \int \left[\frac{\partial}{\partial\theta}\log f_\theta(x_j|x_1,\dots,x_{j-1})\right] f_\theta(x_j|x_1,\dots,x_{j-1})\,d\mu(x_j). \end{aligned}$$

Viewing this integral as an expectation, we see that

$$E_\theta\bigl[Y_j \bigm| X_1,\dots,X_{j-1}\bigr] = 0,$$

which shows that $l_n'(\theta)$, $n \ge 1$, is a martingale. We also assume

$$\begin{aligned} s_j^2 &\stackrel{\text{def}}{=} \operatorname{Var}_\theta\bigl(Y_j \bigm| X_1,\dots,X_{j-1}\bigr) \\ &= -E_\theta\Bigl[\frac{\partial^2}{\partial\theta^2}\log f_\theta(X_j|X_1,\dots,X_{j-1}) \Bigm| X_1,\dots,X_{j-1}\Bigr], \end{aligned}$$

which holds if a second derivative can be passed inside the integral above. By smoothing, $\sigma_j^2 = E_\theta s_j^2 = \operatorname{Var}_\theta(Y_j)$, converging to $I(\theta)$ as $j \to \infty$ by an argument like that for (9.22). Therefore if the Lindeberg condition holds, then

$$\frac{1}{\sqrt{n}}l_n'(\theta) \Rightarrow N\bigl(0, I(\theta)\bigr)$$

as $n \to \infty$ by the martingale central limit theorem. Thus with suitable regularity (9.21) should hold in this setting as it did with i.i.d. observations. Then using (9.20)

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N\big(0, 1/I(\theta)\big)$$

as $n \to \infty$.

The derivation above is sketchy, but can be made rigorous with suitable regularity. Some possibilities are explored in the problems, but good conditions may also depend on the context. Martingale limit theory is developed in Hall and Heyde (1980). The martingale structure of the score function does not depend on stationarity or ergodicity, and Hall and Heyde's book has a chapter on large-sample theory for the maximum likelihood estimator without these restrictions. Results for stationary ergodic Markov chains are given in Billingsley (1961) and Roussas (1972).

9.10 Problems[8]

1. Let $Z_1, Z_2, \dots$ be i.i.d. standard normal, and define random functions G_n, $n \geq 1$, taking values in $\mathcal{C}(K)$ with $K = [0,1]$ by

$$G_n(t) = nZ_n(1-t)t^n - t, \qquad t \in [0,1].$$

   Finally, take $g(t) = EG_n(t) = -t$.
   a) Show that for any $t \in [0,1]$, $G_n(t) \xrightarrow{p} g(t)$.
   b) Compute $\sup_{t\in[0,1]} n(1-t)t^n$ and find its limit as $n \to \infty$.
   c) Show that $\|G_n - g\|_\infty$ does not converge in probability to zero as $n \to \infty$.
   d) Let T_n maximize G_n over $[0,1]$. Show that T_n does not converge to zero in probability.
2. *Method of moments estimation.* Let $X_1, X_2, \dots$ be i.i.d. observations from some family of distributions indexed by $\theta \in \Omega \subset \mathbb{R}$. Let $\overline{X}_n$ denote the average of the first n observations, and let $\mu(\theta) = E_\theta X_i$ and $\sigma^2(\theta) = \mathrm{Var}_\theta(X_i)$. Assume that μ is strictly monotonic and continuously differentiable. The method of moments estimator $\hat{\theta}_n$ solves $\mu(\theta) = \overline{X}_n$. If $\mu'(\theta) \neq 0$, find the limiting distribution for $\sqrt{n}(\hat{\theta}_n - \theta)$.
3. Take $K = [0,1]$, let W_n, $n \geq 1$, be random functions taking values in $\mathcal{C}(K)$, and let f be a constant function in $\mathcal{C}(K)$. Consider the following conjecture. If $\|W_n - f\|_\infty \xrightarrow{p} 0$ as $n \to \infty$, then $\int_0^1 W_n(t)\,dt \xrightarrow{p} \int_0^1 f(t)\,dt$. Is this conjecture true or false? If true, give a proof; if false, find a counterexample.
4.

[8] Solutions to the starred problems are given at the back of the book.

5. maximum likelihood estimation!inconsistent example If $Z \sim N(\mu, \sigma^2)$, then $X = e^Z$ has the lognormal distribution with parameters μ and σ^2. In some situations a threshold γ, included by taking

$$X = \gamma + e^Z,$$

may be desirable, and in this case X is said to have the three-parameter lognormal distribution with parameters γ, μ, and σ^2. Let data $X_1, \dots, X_n$ be i.i.d. from this three-parameter lognormal distribution.

   a) Find the common marginal density for the X_i.
   b) Suppose the threshold γ is known. Find the maximum likelihood estimators $\hat{\mu} = \hat{\mu}(\gamma)$ and $\hat{\sigma}^2 = \hat{\sigma}^2(\gamma)$ of μ and σ^2. (Assume $\gamma < X_{(1)}$.)
   c) Let $l(\gamma, \mu, \sigma^2)$ denote the log-likelihood function. The maximum likelihood estimator for γ, if it exists, will maximize $l\bigl(\gamma, \hat{\mu}(\gamma), \hat{\sigma}^2(\gamma)\bigr)$ over γ. Determine

$$\lim_{\gamma \uparrow X_{(1)}} l\bigl(\gamma, \hat{\mu}(\gamma), \hat{\sigma}^2(\gamma)\bigr).$$

   Hint: Show first that as $\gamma \uparrow X_{(1)}$,

$$\hat{\mu}(\gamma) \sim \frac{1}{n} \log(X_{(1)} - \gamma) \text{ and } \hat{\sigma}^2(\gamma) \sim \frac{n-1}{n^2} \log^2(X_{(1)} - \gamma).$$

   Remark: This thought-provoking example is considered in Hill (1963).

6. Let $X_1, X_2, \dots$ be i.i.d. from a uniform distribution on $(0, 1)$, and let $T_n \in [0, 1]$ be the unique solution of the equation

$$\sum_{i=1}^{n} t^{X_i} = \sum_{i=1}^{n} X_i^2.$$

   a) Show that $T_n \xrightarrow{p} c$ as $n \to \infty$, identifying the constant c.
   b) Find the limiting distribution for $\sqrt{n}(T_n - c)$ as $n \to \infty$.

7. Let $X_1, X_2, \dots$ be i.i.d. from a uniform distribution on $(0, 1)$ and let T_n maximize

$$\sum_{i=1}^{n} \frac{\log(1 + t^2 X_i)}{t}$$

over $t > 0$.

   a) Show that $T_n \xrightarrow{p} c$ as $n \to \infty$, identifying the constant c.
   b) Find the limiting distribution for $\sqrt{n}(T_n - c)$ as $n \to \infty$.

*8. If V and W are independent variables with $V \sim \chi_j^2$ and $W \sim \chi_k^2$, then the ratio $(V/j)/(W/k)$ has an F distribution with j and k degrees of freedom. Suppose $X_1, \dots, X_m$ is a random sample from $N(\mu_x, \sigma_x^2)$ and $Y_1, \dots, Y_n$ is an independent random sample from $N(\mu_y, \sigma_y^2)$. Find a pivotal quantity with an F distribution. Use this quantity to set a $1 - \alpha$ confidence interval for the ratio σ_x/σ_y.

*9. Let $X_1, \dots, X_n$ be i.i.d. from a uniform distribution on $(0, \theta)$.

a) Find the maximum likelihood estimator $\hat{\theta}$ of θ.
b) Show that $\hat{\theta}/\theta$ is a pivotal quantity and use it to set a $1-\alpha$ confidence interval for θ.

*10. Let $X_1, \dots, X_n$ be i.i.d. exponential variables with failure rate θ. Then $T = X_1 + \cdots + X_n$ is complete sufficient. Determine the density of θT, showing that it is a pivot. Use this pivot to derive a $1 - \alpha$ confidence interval for θ.

*11. Consider a location/scale family of distributions with densities $f_{\theta,\sigma}$ given by

$$f_{\theta,\sigma}(x) = \frac{g\left(\frac{x-\theta}{\sigma}\right)}{\sigma}, \qquad x \in \mathbb{R},$$

where g is a known probability density.
a) Find the density of $(X - \theta)/\sigma$ if X has density $f_{\theta,\sigma}$.
b) If X_1 and X_2 are independent variables with the same distribution from this family, show that

$$W = \frac{X_1 + X_2 - 2\theta}{|X_1 - X_2|}$$

is a pivot.
c) Derive a confidence interval for θ using the pivot from part (b).
d) Give a confidence interval for σ based on an appropriate pivot.

*12. Suppose $S_1(X)$ and $S_2(X)$ are both $1-\alpha$ confidence regions for the same parameter $g(\theta)$. Show that the intersection $S_1(X) \cap S_2(X)$ is a confidence region with coverage probability at least $1 - 2\alpha$.

*13. Let (X_1, Y_1), ..., (X_n, Y_n) be i.i.d. with $X_i \sim N(0, 1)$ and $Y_i | X_i = x \sim N(x\theta, 1)$.
a) Find the maximum likelihood estimate $\hat{\theta}$ of θ.
b) Find the Fisher information $I(\theta)$ for a single observation (X_i, Y_i).
c) Determine the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$.
d) Give a $1 - \alpha$ asymptotic confidence interval for θ based on $I(\hat{\theta})$.
e) Compare the interval in part (d) with a $1 - \alpha$ asymptotic confidence interval based on observed Fisher information.
f) Determine the (exact) distribution of $\sqrt{\sum X_i^2}(\hat{\theta} - \theta)$ and use it to find the true coverage probability for the interval in part (e). Hint: Condition on $X_1, \dots, X_n$ and use smoothing.

14. Let $X_1, \dots, X_n$ be a random sample from $N(\theta, \theta^2)$. Give or describe four asymptotic confidence intervals for θ.

*15. Suppose X has a binomial distribution with n trials and success probability p. Give or describe four asymptotic confidence intervals or regions for p. Find these regions numerically if $1 - \alpha = 95\%$, $n = 100$, and $X = 30$.

16. Let $X_1, \dots, X_n$ be i.i.d. from a geometric distribution with success probability p. Describe four asymptotic confidence regions for p.

17. *A variance stabilizing approach.* Let $X_1, X_2, \dots$ be i.i.d. from a Poisson distribution with mean θ, and let $\hat{\theta}_n = \overline{X}_n$ be the maximum likelihood estimator of θ.
    a) Find a function $g : [0, \infty) \to \mathbb{R}$ such that
    $$Z_n = \sqrt{n}\big[g(\hat{\theta}_n) - g(\theta)\big] \Rightarrow N(0, 1).$$
    b) Find a $1 - \alpha$ asymptotic confidence interval for θ based on the approximate pivot Z_n.
18. Let $X_1, X_2, \dots$ be i.i.d. from $N(\mu, \sigma^2)$. Suppose we know that σ is a known function of μ, $\sigma = g(\mu)$. Let $\hat{\mu}_n$ denote the maximum likelihood estimator for μ under this assumption, based on $X_1, \dots, X_n$.
    a) Give a $1-\alpha$ asymptotic confidence interval for μ centered at $\hat{\mu}_n$. Hint: If $Z \sim N(0, 1)$, then $\mathrm{Var}(Z^2) = 2$ and $\mathrm{Cov}(Z, Z^2) = 0$.
    b) Compare the width of the asymptotic confidence interval in part (a) with the width of the t-confidence interval that would be appropriate if μ and σ were not functionally related. Specifically, show that the ratio of the two widths converges in probability as $n \to \infty$, identifying the limiting value. (The limit should be a function of μ.)
19. Suppose that the density for our data X comes from an exponential family with density
    $$h(x)e^{\eta(\theta)T(x) - B(\theta)}, \qquad \theta \in \Omega \subset \mathbb{R}.$$
    If $\hat{\theta}$ is the maximum likelihood estimator of θ, show that $-l''(\hat{\theta})$ and $I(\hat{\theta})$ agree. (Assume that η is differentiable and monotonic.) So in this case, the asymptotic confidence intervals (9.5) and (9.7) are the same.

*20. Suppose electronic components are independent and work properly with probability p, and that components are tested successively until one fails. Let X_1 denote the number that work properly. In addition, suppose devices are constructed using two components connected in series. For proper performance, both components need to work properly, and these devices will work properly with probability p^2. Assume these devices are made with different components and are also tested successively until one fails, and let X_2 denote the number of devices that work properly.
    a) Determine the maximum likelihood estimator of p based on X_1 and X_2.
    b) Give the EM algorithm to estimate p from $Y = X_1 + X_2$.
    c) If $Y = 5$ and the initial guess for p is $\hat{p}_0 = 1/2$, give the next two estimates, $\hat{p}_1$ and $\hat{p}_2$, from the EM algorithm.

*21. Suppose $X_1, \dots, X_n$ are i.i.d. with common (Lebesgue) density
    $$f_\theta(x) = \frac{\theta e^{\theta x}}{2\sinh\theta}, \qquad x \in (-1, 1),$$
    and let $Y_i = I\{X_i > 0\}$. If $\theta = 0$ the X_i are uniformly distributed on $(-1, 1)$.

a) Give an equation for the maximum likelihood estimator $\hat{\theta}_x$ based on $X_1, \dots, X_n$.
b) Find the maximum likelihood estimator $\hat{\theta}_y$ based on $Y_1, \dots, Y_n$.
c) Determine the EM algorithm to compute $\hat{\theta}_y$.
d) Show directly that $\hat{\theta}_y$ is a fixed point for the EM algorithm.
e) Give the first two iterates, $\hat{\theta}_1$ and $\hat{\theta}_2$, of the EM algorithm if the initial guess is $\hat{\theta}_0 = 0$ and there are 5 observations with $Y_1 + \cdots + Y_5 = 3$.

22. Consider a multinomial model for a two-way contingency table with independence, so that $N = (N_{11}, N_{12}, N_{21}, N_{22})$ is multinomial with n trials and success probabilities $\bigl(pq, p(1-q), q(1-p), (1-p)(1-q)\bigr)$. Here $p \in (0,1)$ and $q \in (0,1)$ are unknown parameters.
   a) Find the maximum likelihood estimators of p and q based on N.
   b) Suppose we misplace the off-diagonal entries of the table, so our observed data are $X_1 = N_{11}$ and $X_2 = N_{22}$. Describe in detail the EM algorithm used to compute the maximum likelihood estimators of p and q based on X_1 and X_2.
   c) If the initial guess for p is $2/3$, the initial guess for q is $1/3$, the number of trials is $n = 12$, $X_1 = 4$, and $X_2 = 2$, what are the revised estimates for p after one and two complete iterations of the EM algorithm?
23. Let $X_1, \dots, X_n$ be i.i.d. exponential variables with failure rate λ. Also, for $i = 1, \dots, n$, let $Y_i = I\{X_i > c_i\}$, where the thresholds $c_1, \dots, c_n$ are known constants in $(0, \infty)$.
   a) Derive the EM recursion to compute the maximum likelihood estimator of λ based on $Y_1, \dots, Y_n$.
   b) Give the first two iterates, $\hat{\lambda}_1$ and $\hat{\lambda}_2$, if the initial guess is $\hat{\lambda}_0 = 1$ and there are three observations, $Y_1 = 1$, $Y_2 = 1$, and $Y_3 = 0$, with $c_1 = 1$, $c_2 = 2$, and $c_3 = 3$.
24. *Contingency tables with missing data.* Counts indicating responses to two binary questions, A and B, in a survey are commonly presented in a two-by-two contingency table. In practice, some respondents may only answer one of the questions. If m respondents answer both questions, then cross-classified counts $N = (N_{11}, N_{12}, N_{21}, N_{22})$ for these respondents would be observed, and would commonly be modeled as having a multinomial distribution with m trials and success probability $p = (p_{11}, p_{12}, p_{21}, p_{22})$. Count information for the n_A respondents that only answer question A could be summarized by a variable R representing the number of these respondents who gave the first answer to question A. Under the "missing at random" assumption that population proportions for these individuals are the same as proportions for individuals who answer both questions, R would have a binomial distribution with success probability $p_{1+} = p_{11} + p_{12}$. Similarly, for the n_B respondents who only answer question B, the variable C counting the number who give the first answer to question B would have a binomial distribution with success probability $p_{+1} = p_{11} + p_{21}$.

a) Develop an EM algorithm to find the maximum likelihood estimator of p from these data, N, R, C. The complete data X should be three independent tables N, N^A, and N^B, with sample sizes m, n_A, and n_B, respectively, and common success probability p, related to the observed data by $R = N^A_{1+}$ and $C = N^B_{+1}$.

b) Suppose the observed data are

$$N = \begin{pmatrix} 5 & 10 \\ 10 & 5 \end{pmatrix}, \qquad R = 5, \qquad C = 10,$$

with $m = 30$ and $n_A = n_B = 15$. If the initial guess for p is $\hat{p}_0 = N/30$, find the first two iterates for the EM algorithm, $\hat{p}_1$ and $\hat{p}_2$.

25. *A simple hidden Markov model.* Let $X_1, X_2, \ldots$ be Bernoulli variables with $EX_1 = 1/2$ and the joint mass function determined recursively by

$$P(X_{k+1} \neq x_k | X_1 = x_1, \ldots, X_k = x_k) = \theta, \qquad n = 1, 2, \ldots.$$

Viewed as a process in time, X_n, $n \geq 1$, is a Markov chain on $\{0, 1\}$ that changes state at each stage with probability θ. Suppose these variables are measured with error. Specifically, let $Y_1, Y_2, \ldots$ be Bernoulli variables that are conditionally independent given the X_i, satisfying

$$P(Y_i \neq X_i | X_1, X_2, \ldots) = \gamma.$$

Assume that the error probability γ is a known constant, and $\theta \in (0, 1)$ is an unknown parameter.

a) Show that the joint mass functions for $X_1, \ldots, X_n$ form an exponential family.

b) Find the maximum likelihood estimator for θ based on $X_1, \ldots, X_n$.

c) Give formulas for the EM algorithm to compute the maximum likelihood estimator of θ based on $Y_1, \ldots, Y_n$.

d) Give the first two iterates for the EM algorithm, $\hat{\theta}_1$ and $\hat{\theta}_2$, if the initial guess is $\hat{\theta}_0 = 1/2$, the error probability γ is 10%, and there are four observations: $Y_1 = 1$, $Y_2 = 1$, $Y_3 = 0$ and $Y_4 = 1$.

26. *Probit analysis.* Let $Y_1, \ldots, Y_n$ be independent Bernoulli variables with

$$P(Y_i = 1) = \Phi(\alpha + \beta t_i),$$

where $t_1, \ldots, t_n$ are known constants and α and β in $\mathbb{R}$ are unknown parameters. Also, let $X_1, \ldots, X_n$ be independent with $X_i \sim N(\alpha + \beta t_i, 1)$, $i = 1, \ldots, n$.

a) Describe a function $g : R^n \to \{0, 1\}^n$ such that $Y \sim g(X)$ for any α and β.

b) Find the maximum likelihood estimator for $\theta = (\alpha, \beta)$ based on X.

c) Give formulas for the EM algorithm to compute the maximum likelihood estimator of θ based on Y.

d) Suppose we have five observations, $Y = (0, 0, 1, 0, 1)$ and $t_i = i$ for $i = 1, \dots, 5$. Give the first two iterates for the EM algorithm, $\hat{\theta}_1$ and $\hat{\theta}_2$, if the initial guess is $\hat{\theta}_0 = (-2, 1)$.

*27. Suppose $X_1, X_2, \dots$ are i.i.d. with common density f_θ, where $\theta = (\eta, \lambda) \in \Omega \subset \mathbb{R}^2$. Let $I = I(\theta)$ denote the Fisher information matrix for the family, and let $\hat{\theta}_n = (\hat{\eta}_n, \hat{\lambda}_n)$ denote the maximum likelihood estimator from the first n observations.

a) Show that $\sqrt{n}(\hat{\eta}_n - \eta) \Rightarrow N(0, \tau^2)$ under P_θ as $n \to \infty$, giving an explicit formula for τ^2 in terms of the Fisher information matrix I.

b) Let $\tilde{\eta}_n$ denote the maximum likelihood estimator of η from the first n observations when λ has a known value. Then $\sqrt{n}(\tilde{\eta}_n - \eta) \Rightarrow N(0, \nu^2)$ under P_θ as $n \to \infty$. Give an explicit formula for ν^2 in terms of the Fisher information matrix I, and show that $\nu^2 \le \tau^2$. When is $\nu^2 = \tau^2$?

c) Assume $I(\cdot)$ is a continuous function, and derive a $1 - \alpha$ asymptotic confidence interval for η based on the plug-in estimator $I(\hat{\theta}_n)$ of $I(\theta)$.

d) The observed Fisher information matrix for a model with several parameters can be defined as $-\nabla^2 l(\hat{\theta}_n)$, where ∇^2 is the Hessian matrix of partial derivatives (with respect to η and λ). Derive a $1 - \alpha$ asymptotic confidence interval for η based on observed Fisher information instead of $I(\hat{\theta}_n)$.

28. Let $(X_1, Y_1), (X_2, Y_2), \dots$ be i.i.d. with common Lebesgue density

$$\frac{\exp\left\{-\dfrac{(x-\mu_x)^2 - 2\rho(x-\mu_x)(y-\mu_y) + (y-\mu_y)^2}{2(1-\rho^2)}\right\}}{2\pi\sqrt{1-\rho^2}},$$

where $\theta = (\mu_x, \mu_y, \rho) \in \mathbb{R}^2 \times (-1, 1)$ is an unknown parameter. (This is a bivariate normal density with both variances equal to one.)

a) Give formulas for the maximum likelihood estimators of μ_x, μ_y, and ρ.

b) Find the (3×3) Fisher information matrix $I(\theta)$ for a single observation.

c) Derive asymptotic confidence intervals for μ_x and ρ based on $I(\hat{\theta})$ and based on observed Fisher information (so you should give four intervals, two for μ_x and two for ρ).

*29. Suppose W and X have a known joint density q, and that

$$Y | W = w,\ X = x \sim N(\alpha w + \beta x, 1).$$

Let $(W_1, X_1, Y_1), \dots, (W_n, X_n, Y_n)$ be i.i.d., each with the same joint distribution as (W, X, Y).

a) Find the maximum likelihood estimators $\hat{\alpha}$ and $\hat{\beta}$ of α and β. Determine the limiting distribution of $\sqrt{n}(\hat{\alpha} - \alpha)$. (The answer will depend on moments of W and X.)

b) Suppose β is known. What is the maximum likelihood estimator $\tilde{\alpha}$ of α? Find the limiting distribution of $\sqrt{n}(\tilde{\alpha} - \alpha)$. When will the limiting distribution for $\sqrt{n}(\tilde{\alpha} - \alpha)$ be the same as the limiting distribution in part (a)?

*30. Prove Proposition 9.31 and show that (9.13) is a $1 - \alpha$ asymptotic confidence interval for $g(\theta)$. Suggest two estimators for $\nu(\theta)$.

*31. Let $N_{11}, \dots, N_{22}$ be cell counts for a two-way table with independence. Specifically, N has a multinomial distribution on n trials, and the cell probabilities satisfy $p_{ij} = p_{i+}p_{+j}$, $i = 1, 2$, $j = 1, 2$. The distribution of N is determined by $\theta = (p_{+1}, p_{1+})$. Find the maximum likelihood estimator $\hat{\theta}$, and give an asymptotic confidence intervals for $p_{11} = \theta_1\theta_2$.

32. Suppose $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. random vectors in $\mathbb{R}^2$ with common density

$$\frac{\exp\bigl\{-(x-\theta_x)^2 + \sqrt{2}(x-\theta_x)(y-\theta_y) - (y-\theta_y)^2\bigr\}}{\pi\sqrt{2}}.$$

In polar coordinates we can write $\theta_x = \|\theta\| \cos\omega$ and $\theta_y = \|\theta\| \sin\omega$, with $\omega \in (-\pi, \pi]$. Derive asymptotic confidence intervals for $\|\theta\|$ and ω.

33. Suppose $X_1, X_2, \dots$ are i.i.d. from some distribution Q_θ, with $\theta \in \Omega \subset \mathbb{R}^p$. Assume that the Fisher information matrix $I(\theta)$ exists and is positive definite and continuous as a function of θ. Also, assume that the family $\{Q_\theta : \theta \in \Omega\}$ is regular enough that the maximum likelihood estimators $\hat{\theta}_n$ are consistent, and that

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N\bigl(0, I(\theta)^{-1}\bigr).$$

a) Find the limiting distribution for

$$\sqrt{n}I(\theta)^{1/2}(\hat{\theta}_n - \theta).$$

b) Find the limiting distribution for

$$n(\hat{\theta}_n - \theta)'I(\hat{\theta})(\hat{\theta}_n - \theta).$$

Hint: This variable should almost be a function of the random vector in part (a).

c) The variable in part (b) should be an asymptotic pivot. Use this pivot to find an asymptotic $1 - \alpha$ confidence region for θ. (Use the upper quantile for the limiting distribution only.)

d) If $p = 2$ and $I(\theta)$ is diagonal, describe the shape of your asymptotic confidence region. What is the shape of the region if $I(\theta)$ is not diagonal?

34. *Simultaneous confidence intervals.* Suppose $X_1, X_2, \dots$ are i.i.d. from some distribution Q_θ with θ two-dimensional:

$$\theta = \begin{pmatrix} \beta \\ \lambda \end{pmatrix} \in \Omega \subset \mathbb{R}^2.$$

Assume that the Fisher information matrix $I(\theta)$ exists, is positive definite, is a continuous function of θ, and is diagonal,

$$I(\theta) = \begin{pmatrix} I_\beta(\theta) & 0 \\ 0 & I_\lambda(\theta) \end{pmatrix}.$$

Finally, assume the family $\{Q_\theta : \theta \in \Omega\}$ is regular enough that the maximum likelihood estimators are asymptotically normal:

$$\sqrt{n}(\hat{\theta}_n - \theta) = \sqrt{n}\left(\begin{pmatrix} \hat{\beta}_n \\ \hat{\lambda}_n \end{pmatrix} - \begin{pmatrix} \beta \\ \lambda \end{pmatrix}\right) \Rightarrow N\bigl(0, I(\theta)^{-1}\bigr).$$

a) Let

$$M_n = \max\left\{\sqrt{nI_\beta(\hat{\theta}_n)}\,|\hat{\beta}_n - \beta|, \sqrt{nI_\lambda(\hat{\theta}_n)}\,|\hat{\lambda}_n - \lambda|\right\}.$$

Show that $M_n \Rightarrow M$ as $n \to \infty$. Does the distribution of M depend on θ?

b) Let q denote the upper αth quantile for M. Derive a formula relating q to quantiles for the standard normal distribution.

c) Use M_n to find a $1 - \alpha$ asymptotic confidence region S for θ. (You should only use the upper quantile for the limiting distribution.) Describe the shape of the confidence region S.

d) Find intervals CI_β and CI_λ based on the data, such that

$$P(\beta \in CI_\beta \text{ and } \lambda \in CI_\lambda) \to 1 - \alpha.$$

From this, it is natural to call intervals CI_β and CI_λ asymptotic simultaneous confidence intervals for λ and β, because the chance they simultaneously cover β and λ is approximately $1 - \alpha$.

35. *Multivariate confidence regions.* Let $X_1, \dots, X_m$ be a random sample from $N(\mu_x, 1)$ and $Y_1, \dots, Y_n$ be a random sample from $N(\mu_y, 1)$, with all $m+n$ variables independent, and let $\overline{X} = (X_1 + \cdots + X_m)/m$ and $\overline{Y} = (Y_1 + \cdots + Y_n)/n$.

a) Find the cumulative distribution function for

$$V = \max\bigl\{|\overline{X} - \mu_x|, |\overline{Y} - \mu_y|\bigr\}.$$

b) Assume $n = m$ and use the pivot from part (a) to find a $1 - \alpha$ confidence region for $\theta = (\mu_x, \mu_y)$. What is the shape of this region?

36. Let $X_1, X_2, \dots$ be i.i.d. from a distribution Q that is symmetric about θ. Let $\overline{X}_n$ and $\tilde{X}_n$ denote the mean and median of the first n observations, and let T_n be the M-estimator from the first n observations using the function ρ given in Theorem 9.34 with $k = 1$.

a) Determine the asymptotic relative efficiency of T_n with respect to $\overline{X}_n$ if $Q = N(\theta, 1)$.

b) Determine the asymptotic relative efficiency of T_n with respect to $\tilde{X}_n$ if Q is absolutely continuous with density

$$\frac{1}{\pi[(x-\theta)^2+1]},$$

a Cauchy density with location θ.

37. Let $X_1, X_2, \dots$ be i.i.d. from a distribution Q that is symmetric about θ and absolutely continuous with density q. Fix ϵ (or k), and let T_n be the M-estimator from the first n observations using the function ρ given in Theorem 9.34. Take $\psi = \rho'$.
   a) Suggest a consistent estimator for $\lambda'(\theta)$. You can assume that $\lambda'(\theta) = -E\psi'(X-\theta)$.
   b) Suggest a consistent estimator for $E\psi^2(X-\theta)$.
   c) Using the estimators in parts (a) and (b), find an asymptotic $1-\alpha$ confidence interval for θ.
38. Suppose $Y \sim N(0,1)$ and that, given $Y = y$, X_n, $n \in \mathbb{Z}$, are i.i.d. from $N(y,1)$. Find a shift invariant set B with $P(X \in B) = 1/2$.
39. Show that if $E|Y_i|^{2+\epsilon}$ is bounded for some $\epsilon > 0$, then the Lindeberg condition (9.17) holds.
40. Show that if

$$E_\theta\left(g(x) - \log\frac{f_\theta(X_0|X_{-1},\dots,X_{-k})}{f_\omega(X_0|X_{-1},\dots,X_{-k})}\right)^2, \qquad k = 1, 2, \dots,$$

   are finite[9] and tend to zero as $k \to \infty$, then the approximation error in (9.19) tends to zero in probability as $n \to \infty$.
41. Let X_n, $n \in \mathbb{Z}$, be a stationary process with X_0 uniformly distributed on $(0,1)$, satisfying

$$X_n = \langle 2X_{n+1}\rangle, \qquad n \in \mathbb{Z}.$$

   Here $\langle x\rangle \overset{\text{def}}{=} x - \lfloor x\rfloor$ denotes the fractional part of x. Show that $X - 1/2$ is a linear process. Identify a distribution Q for the innovations ϵ_i and coefficients c_n, $n \geq 1$.

[9] Actually, it is not hard to argue that the conclusion will still hold if some of these moments are infinite, provided they still converge to zero.